



Rent • Manage • Maintain • Sell

High-Level Product Requirements & User Story Catalogue

A mobile-first, one-stop property operating system for India — covering residential rentals, hostels and PG accommodation, and commercial premises, with the property manager as the operating middle layer between owners, tenants and service providers. Free to use, monetised on transactions.

Document Product Requirements — High Level (v2.0)

Product dwellm8 · `tesseract/dwellm8`

Date 29 July 2026

Monetisation No subscriptions — 2.99% platform fee deducted by the manager at payout; instrument costs passed through; premium AI tier ₹499/org/month (future)

Status Draft for review — supersedes the vertical and pricing scope in `docs/product-brief.md`

Audience Product, engineering, design, compliance and commercial stakeholders

§ Contents

1. **Executive summary** — the one-stop thesis, and what is deliberately not in it
2. **Scope at a glance** — verticals × personas × modules
3. **Personas and roles** — nine actors, what each is accountable for
4. **The three verticals** — residential, hostel/PG/co-living, commercial
5. **The operating model** — money flows, cost sharing, the manager's mandate
6. **Module requirements and user stories** — M1–M20, 296 stories
7. **Mobile-first experience requirements** — five apps and the web surfaces; Admin is both
8. **Critical end-to-end workflows** — the five journeys that cross every module
9. **Integrations, data, stack, identity and access** — adapters, stack, the authorisation model, access tiers and demo data
10. **Non-functional requirements** — reliability, security, performance, scale
11. **India regulatory summary** — payments, documents, tax, privacy
12. **Phased delivery plan** — what ships when, and why in that order
13. **Assumptions and open decisions** — what must be answered before build
14. **Appendix — competitive parity check** — PropertyMe comparison, gaps found and closed

1. Executive summary

dwelIm8 is the system of record for a property in India — one property, one unit, one occupancy, one ledger, one document set — with a front door for each party who touches it: the owner, the property manager, the occupant, the service provider and the platform operator. It runs three verticals on one core, and it extends from renting a property to maintaining it and ultimately to selling it.

1.1 The problem

Indian property is operated on WhatsApp threads, Excel files and cash. An owner does not know what was collected, what was spent, or what is owed. A manager runs a book of hundreds of units across a dozen spreadsheets and a personal phone. A tenant has no record of anything, and finds an electrician through a neighbour. A hostel operator tracks beds on a whiteboard. A commercial landlord reconciles CAM once a year and loses the argument. Every one of these parties knows exactly what they are owed and can prove none of it.

1.2 The thesis

Three products are usually built separately and sold separately: **rent collection software**, **facilities and maintenance software**, and **a services marketplace**. Each on its own is a thin, low-retention business. Together, with the property manager as the operating layer between them, they compound:

- Collections give the platform **the money rail and the ledger**, so any cost can be recovered from the party who owes it without anyone chasing a transfer.
- Maintenance gives the platform **the recurring reason to open the app** between rent cycles, which collections alone never produce.
- The service marketplace turns maintenance from a cost centre into **transaction revenue**, and gives the manager a supply base they do not have to build.
- The manager is the party who **brings units at scale** — a single firm onboards three hundred units and, with them, three hundred tenants and ninety owners.

1.3 What "one-stop shop" means concretely

Collect

Rent, hostel fees, society dues and CAM billed automatically and collected over UPI, autopay mandates, NACH, cards or recorded offline. Receipts, reminders and reconciliation are part of it.

Maintain

A ticket becomes a quote, an approval, a dispatched technician, a completed job with photographic evidence, an invoice, a ledger posting against whoever is liable, and a warranty.

Comply

Agreements drafted, stamped, e-signed and stored. KYC, police verification, GST, TDS and statutory notices tracked as workflow rather than advice.

Operate

Bed allocation for hostels, CAM and meter billing for commercial, visitor and gate management for buildings, staff and vendor management for managers.

Settle

Owner payouts net of fee, GST, TDS and shared costs, with a statement in which every line traces to a ledger posting to the paisa.

Sell

The same unit, tenant history and document vault becomes a verified listing, a lead pipeline, a site-visit log, a token advance and a registered sale.

1.4 What makes it defensible

1. **Money is a double-entry ledger, not a status column.** Rent, fees, deposits, shared costs, vendor bills, payouts and refunds are postings. A balance is always derived. This is what lets four parties look at the same job and agree on who

owes what.

2. **Cost-sharing is a rule engine, not a conversation.** Who pays for a repair is decided by the agreement before the technician is called, not argued after the invoice arrives. This single mechanism is the reason the manager, the owner and the tenant can all trust the same system.
3. **Compliance artefacts are first-class objects.** An agreement knows its stamp duty, its e-sign audit trail and its expiry. A vendor invoice knows its GST and its TDS. A PDF in a folder is not a record.
4. **One core, three verticals.** A hostel bed, a 2BHK and a 4,000 sq ft office are the same unit object with different billing bases and operating modules. Competitors build three products; the third one never ships.

1.5 Deliberate non-goals

OUT OF SCOPE — PERMANENTLY OR UNTIL EXPLICITLY REVISITED

Not a lender. Rent-now-pay-later, deposit financing and home loans are partner-referred, never balance-sheet products.

Not an escrow or fund custodian. Money settles owner-side through a regulated aggregator. Any design in which dwellm8 holds rent overnight in a platform account must be escalated, not implemented.

Not an employer of technicians. Service providers are onboarded partners with their own compliance obligations; the platform must not drift into a de facto employment relationship.

Not a tenant credit bureau. Screening output is advisory to the requesting owner. There is no portable tenant score and no cross-owner blacklist.

Not a developer sales channel at launch. New-project primary sales is a later, separate motion from resale and rental.

2. Scope at a glance

2.1 The scope cube

Every requirement in this document sits at the intersection of a **vertical** (what kind of property), a **persona** (who is acting) and a **module** (what capability). The matrix below is the map of the product.

Module	Residential	Hostel / PG	Commercial	Primary personas
M1 Identity & organisations	Core	Core	Core	All
M2 Property & unit inventory	Unit	Room + bed	Floor / suite	Owner, Manager
M3 Leasing & agreements	Lease	Licence	Registered lease	Owner, Manager, Tenant
M4 Occupant onboarding & KYC	Core	+ guardian, college	+ entity KYC	Tenant, Manager
M5 Billing & collections	Rent	Fee + mess	Rent + CAM + meters	Tenant, Manager
M6 Deposits & settlement	Core	Caution deposit	6–12 month deposit	Tenant, Owner
M7 Maintenance & ticketing	Core	Core	+ AMC, PPM	Tenant, Manager
M8 Service provider marketplace	Core	Core	Core	Vendor, Manager
M9 Cost sharing & approvals	Core	Core	Core	Owner, Manager, Tenant
M10 Accounting, payouts & tax	Core	Core	+ RCM, 194-I	Owner, Manager, Admin
M11 Hostel operations	—	Core	—	Warden, Resident, Guardian
M12 Commercial operations	—	—	Core	Manager, Tenant entity
M13 Communication & notifications	Core	Core	Core	All
M14 Document vault & compliance	Core	Core	Core	All
M15 Sales & brokerage	Core	Asset sale	Core	Owner, Agent, Buyer
M16 Platform administration	Cross-cutting			Platform Admin
M17 Analytics & reporting	Cross-cutting			Owner, Manager, Admin
M18 AI capabilities (<i>premium</i>)	Cross-cutting — the only paid tier			Manager, Owner, Operator
M19 Demo & sandbox	Cross-cutting — every vertical demonstrable			Visitor, Sales, Prospect
M20 Workflow automation	Cross-cutting			Manager, Operator

2.2 Story priority convention

Every story in section 6 carries a priority. **P0** platform-critical — nothing works correctly without it. **P1** launch core — required for the first paying customer in that vertical. **P2** expansion — needed to scale beyond the first cohort. **P3** later — ecosystem, enterprise or opportunistic.

3. Personas and roles

Nine actors touch the platform. Four of them are paying or potentially paying parties, three are operational, and two are platform-side. Every screen, permission and notification in this document belongs to one of them.

Persona	Who they are	What they are accountable for	Primary surface
Owner	Individual landlord (1–20 units), HNI portfolio holder, or NRI owner managing remotely	Owns the asset, sets the commercial terms, approves spend above the manager's mandate, receives the payout	Owner app (mobile-first) + web for portfolio and tax
Property Manager	Firm or individual managing units they do not own, across residential, hostel and commercial	Day-to-day operations, collections, tenant relations, maintenance dispatch, owner reporting — the middle layer	Manager console (web primary) + field app
Field agent	The manager's staff — collections, inspections, move-in/move-out, vendor supervision	Executing tasks on the ground with scoped access to only their assigned properties	Field app (mobile only, offline-tolerant)
Tenant / Resident	Family renting a flat, student or working professional in a hostel/PG, or a business occupying commercial space	Paying on time, raising and verifying maintenance, honouring the agreement, move-out condition	Tenant app (mobile-first, the highest-volume surface)
Guardian	Parent or sponsor of a hostel resident, frequently the actual payer	Paying fees, receiving attendance and leave visibility, emergency contact	Guardian view inside the tenant app
Warden / Hostel operator	Runs the hostel property day to day — beds, mess, attendance, discipline	Occupancy, resident welfare and safety, mess operations, in–out register	Warden console + gate app
Service provider	Vendor firm and its individual technicians — plumber, electrician, carpenter, painter, pest control, AC, cleaning	Accepting jobs, quoting honestly, completing to SLA with evidence, honouring warranty	Vendor app (technician) + vendor web (firm)
Agent / Broker	dwelldm8-side or partner agent handling lease origination and property sales	Listing quality, lead conversion, site visits, negotiation, closing and handover	Agent app + CRM in the manager console
Prospect	Someone looking for a flat, a hostel bed or commercial space — not yet a customer, often not yet a user	Finding a property, contacting the manager, attending an inspection, and applying	Public listing site (web, SEO) + dwelldm8 Live app
Platform Admin	dwelldm8's own operations, support, finance and compliance staff	Onboarding approval, statutory rule tables, dispute desk, reconciliation, platform health	Admin console (web) + Admin app (mobile)

THE IDENTITY RULE THAT EVERYTHING DEPENDS ON

A person is **one identity holding many scoped relationships**. The same phone number is an owner of two flats in Pune, a tenant in Bengaluru, a guardian of a hostel resident in Manipal and a committee member of a society in Noida. Roles are always scoped to (organisation, property, unit) and never global. Permissions are never merged across organisations, and the app switches context explicitly rather than blending it.

4. The three verticals

One core model — organisation, property, unit, occupancy, ledger — carries all three verticals. What differs is the inventory granularity, the billing basis, the agreement instrument and the operating module bolted on top. Building this as three products is the mistake this section exists to prevent.

4.1 What is identical across all three

- Organisation, property and unit hierarchy, with effective-dated ownership
- Occupancy as a contract with parties, a schedule, a deposit and documents
- The double-entry ledger, invoicing, collections, reminders and reconciliation
- Maintenance tickets, cost-sharing rules, vendor dispatch and job costing
- Owner payouts, manager fees, GST and TDS treatment, statements
- Document vault, KYC, notifications and the audit trail

4.2 What differs

Dimension	Residential	Hostel / PG / Co-living	Commercial
Inventory unit	Flat or house	Bed within a room within a block; sharing type drives price	Floor, suite or shop, priced on chargeable area
Instrument	11-month leave & licence agreement	Licence / admission agreement, often with house rules annexure	Registered lease deed; long tenure normal
Billing basis	Monthly rent, fixed	Monthly fee bundling stay + mess + utilities + laundry; admission fee one-time	Rent per sq ft + CAM per sq ft + metered utilities + DG + parking
Deposit	2–3 months (state caps where the Model Tenancy Act is adopted)	Small refundable caution deposit; often one month	6–12 months, sometimes bank guarantee
Payer	Tenant, occasionally employer	Frequently the guardian, not the occupant	A company, with a procurement and PO process
Term mechanics	11 months, renewal with escalation	Term-linked (academic year), high churn, mid-term joining and vacating	Lock-in, rent-free fit-out, escalation every 3 years, exit dilapidation
Tax	Largely exempt residential; §194-IB TDS by individuals	Composite supply questions; GST where services dominate	GST on rent, reverse charge cases, §194-I TDS monthly
Operating module	Maintenance and community	Beds, mess, attendance, leave, curfew, guardian visibility	CAM reconciliation, AMC, planned preventive maintenance, fit-out approvals
Churn	~11–24 months	3–12 months — onboarding and exit must be near-instant	3–9 years
Volume shape	Many owners, few units each	Few operators, hundreds of beds each	Few tenants, very high value each

THE HOSTEL CONSTRAINT THAT SHAPES THE DATA MODEL

A hostel resident occupies a **bed**, not a unit, and moves between beds mid-term. A bed must therefore be a first-class inventory object with its own availability calendar, price and occupancy history — not a text field on a room. Retrofitting bed-level inventory onto a flat-level model is the single most expensive mistake available in this product, and it must be designed in from M2.

THE COMMERCIAL CONSTRAINT THAT SHAPES THE MONEY MODEL

Commercial occupancy generates **several concurrent recurring charges on different bases** — rent on chargeable area, CAM on a budget that is reconciled to actuals annually, electricity on a meter reading, DG on units consumed, parking per bay. The charge engine must support multiple heads with different bases, different escalation rules and an annual true-up, or commercial cannot be billed at all.

5. The operating model — money, mandate and cost sharing

This section is the commercial heart of the product. It defines who pays whom, who may commit whose money, and how a single repair bill is divided between an owner, a tenant and a manager without anybody arguing. Every module in section 6 assumes these rules.

5.1 The four money flows

FLOW A — COLLECTION	Tenant / Guardian → Aggregator → Owner settlement account (rent, fee, CAM, mess, utilities, late fee, recovered costs)
FLOW B — PAYOUT	Collections → [- manager fee - GST - TDS - shared costs] → Owner
FLOW C — SERVICE JOB	Liabe party → Vendor (direct) or Platform pays vendor → recovers from liabe party in the next cycle
FLOW D — PLATFORM REVENUE	2.99% deducted by the manager when paying the owner (outbound leg only; tenant never surcharged) + card/netbanking instrument cost passed through separately + premium AI tier INR 499/org/month (future) + sale brokerage share + partner referral fees

CUSTODY POSTURE — NON-NEGOTIABLE

dwelldm8 is **not an escrow and not a custodian of client funds**. Collections settle to the owner or the manager's regulated account through the payment aggregator. Where the platform advances a vendor payment before recovering it, that advance must be structured, capped, and reviewed against RBI payment-aggregator directions before it is built. A design in which rent accumulates in a dwelldm8-controlled account is escalated, never implemented.

5.2 The manager's mandate — how the middle layer is authorised

The manager acts on the owner's behalf, so the extent of that authority must be an explicit, revocable, effective-dated object rather than an assumption. The **management mandate** is agreed once per owner (or per property) and governs everything the manager may do without asking again.

Mandate term	What it authorises	Typical setting
Scope	Which properties and units the manager may act on	Named units, effective-dated
Fee model	How the manager is paid	% of collections, flat per unit, or per bed; plus origination fee
Spend authority	Maximum a manager may approve on the owner's behalf per job, without owner approval	₹2,000–₹10,000 per job; a monthly cap as well
Emergency authority	Higher ceiling for safety, water, electrical or lift emergencies, with mandatory notification	2–3× the standard ceiling, notify within 1 hour
Maintenance float	Owner-funded working balance the manager may draw on, with a replenishment trigger	₹10,000–₹25,000 per property
Collection authority	Whether the manager may collect into their own account or only into the owner's	Owner account by default

Leasing authority	Whether the manager may sign a lease, and within what rent band	Recommend only, or sign within $\pm 5\%$ of a floor rent
Vendor choice	Whether the manager may use their own vendors or must use platform-verified ones	Platform-verified by default
Revocation	Owner may revoke instantly; access ends immediately, history is retained	Immediate, with in-flight work rules

5.3 The cost-sharing rule engine

Most maintenance disputes are not about the repair; they are about who pays. dwellm8 settles that **before the technician is dispatched** by evaluating a liability matrix carried on the occupancy agreement.

Input	Detail
Category	Plumbing, electrical, carpentry, appliance, structural, pest, painting, common-area, safety
Cause	Fair wear and tear · tenant negligence or misuse · age or defect of asset · external or force majeure
Value band	Threshold rules — for example, tenant bears up to ₹1,000 per event, owner above it
Asset ownership	Owner-provided fixture vs tenant's own appliance
Occupancy age	First 30 days of occupancy commonly owner-borne as a snagging period
Vertical override	Hostel — operator bears almost everything; commercial — tenant bears interiors, owner bears structure and core services

OUTPUTS OF AN EVALUATION

- A **split** — 100% owner, 100% tenant, 100% society/operator, or a percentage or threshold split across two parties
- The **approver** — determined by the split and the manager's spend authority
- The **recovery route** — netted from the owner payout, added to the tenant's next invoice, drawn from the maintenance float, or settled against the deposit at exit
- A **disclosure** shown to both parties, citing the rule, before any spend is committed

WORKED EXAMPLE — THE FLOW THE WHOLE PRODUCT EXISTS TO MAKE BORING

A tenant reports a leaking geyser at 21:40. The matrix reads: category *plumbing*, cause *asset defect*, asset *owner-provided*, value band *above ₹1,000* → **owner-borne**. The manager's mandate allows ₹5,000 per job, so no owner approval is needed. A verified plumber is dispatched for 08:00. The quote comes in at ₹3,200 (₹2,400 labour, ₹800 parts) — within mandate, auto-approved, both parties notified with the rule cited. The job completes with before/after photos and a tenant OTP sign-off. The vendor invoices ₹3,200 + GST; TDS under §194-C is deducted; the platform takes its commission. The cost posts against the *owner*, appears on the owner's monthly statement as a single traceable line, and is netted from the payout. The tenant pays nothing and is never asked to. Total human decisions required: zero.

5.4 Approval ladder

Value	Approver	Behaviour
Below float threshold	Manager, auto-approved	Notify owner, draw on float, no interruption
Within spend authority	Manager	Notify owner with the quote and the liability rule applied
Above spend authority	Owner	Push and WhatsApp with quote, photos and a one-tap approve/decline; escalates on no response
Above owner's own threshold	Owner, with a second quote required	Platform requests comparative quotes before presenting

Tenant-borne, any value	Tenant	Tenant approves and pays, or defers it to deposit settlement with acknowledgement
Emergency	Manager under emergency authority	Proceed immediately, notify within the mandated window, full evidence pack afterwards

5.5 Monetisation — a single transaction fee, no subscriptions

dwelldm8 does **not** sell seats, tiers or per-unit subscriptions. Every organisation — a landlord with one flat, a firm with three hundred units, a hostel with four hundred beds — gets the **complete product** for free and pays only when money moves through it.

THE PRICING RULE

A flat 2.99% platform fee, deducted by the property manager when they pay the owner. It is charged once per rent cycle, on the outbound leg only. The tenant pays exactly the rent — there is no surcharge on money coming in. No subscription, no per-unit charge, no feature gating, no seat limits. Every capability in this document — collections, agreements, maintenance, marketplace, hostel, commercial, accounting, sales — is included from day one. The only future exception is the premium AI tier described in §5.6.

WHY THIS WORKS

- **Zero adoption friction.** The single hardest thing to sell an Indian landlord or hostel operator is a monthly fee for software they have never used. A free product that charges only on success removes the entire objection.
- **Revenue scales with actual usage,** not with seats a customer may never log into. A manager onboarding three hundred units becomes valuable immediately without a procurement conversation.
- **Incentives align.** dwelldm8 earns when collections work. That is exactly what the platform is optimised to do, and it makes reliability a revenue problem rather than a support problem.
- **One number to explain.** No plan comparison table, no upgrade prompts, no feature paywalls in the middle of a workflow.

WHAT THE FEE APPLIES TO

Transaction	Fee applies	Notes
Inbound leg — rent, hostel fee, dues, CAM, utilities paid in-app by the tenant	No	The tenant pays exactly what is due. A surcharge here is what drives money back to cash.
Deposits and advances collected in-app	Yes	Fee is charged once at collection, never again at refund
Service job payments through the marketplace	Yes	Replaces a separate vendor commission — one rate, everywhere
Token advance and brokerage settled in-app	Yes	Sale consideration itself is out of scope — the platform never touches it
Offline cash and direct bank transfers recorded but not transacted	No	Recording stays free on both legs; it is the honesty mechanism that keeps the ledger true
Refunds, reversals and deposit returns	No	Never fee a customer for getting their own money back
Outbound leg — the manager paying the owner	Yes — 2.99% <i>deducted by the manager at payout</i>	The single charge point. The manager is the party dwelldm8 does the operating work for, and this is where the owner statement already itemises deductions.

WHO BEARS IT, AND WHERE IT IS CHARGED

Decided: the property manager bears the fee, and it is deducted at the point they pay the owner. One charge point per rent cycle, on money going out — not on money coming in. This mirrors how an owner already experiences property

management: the statement shows the rent collected, then the deductions, then the net.

Arrangement	Charged	Detail
Property under management	Manager, at payout	Appears as its own line on the owner statement alongside the management fee, so the owner sees exactly what the platform cost
Owner self-managing, no manager	Owner, at collection	There is no payout leg and no manager, so the single charge falls to the owner and is disclosed on their statement
Hostel or co-living	Operator	The operator is the manager in this vertical
Society or RWA dues	Society	The society is the collecting and operating entity
Marketplace service job	Vendor, at settlement	Replaces a separate marketplace commission
Tenant, any payment	Never	The tenant is never surcharged for paying rent through the platform
Refunds, reversals, deposit returns	Never	Nobody is charged for getting their own money back
Offline cash and bank transfer	Never on the inbound leg	Recording stays free; if that money is later paid out through the platform, the payout is charged like any other

PAYMENT INSTRUMENT COSTS ARE SEPARATE AND PASSED THROUGH

The 2.99% is a **platform fee, not a payment-processing charge**. Where a payment instrument carries a real cost — credit card, some netbanking, NACH — that cost is handled separately as a disclosed pass-through surcharge at the point of payment, exactly as Indian rent-payment services already do. UPI and offline transfers carry no instrument cost and therefore no surcharge, which keeps the default path free for the tenant and cheap for the platform.

THE ECONOMICS TO KEEP AN EYE ON

A manager typically earns **8–10% of collected rent**, so 2.99% of gross rent is roughly **30–37% of their revenue** — ₹822 against a ₹2,200 fee on ₹27,500 of rent. That is the first calculation a manager will run, and it is a high share by international norms: Australian property-management platforms charge the agency a flat per-property subscription of roughly A\$2.50–3.25 per property per month all-in, which on a typical rent is on the order of *0.1% of rent* rather than 2.99%.

India is not Australia — there is no statutory trust-account regime, managers are smaller, software willingness-to-pay is far lower, and dwellm8 bundles collections, maintenance, a vendor marketplace and compliance rather than selling a ledger. A transaction model may well be the right answer here where a subscription is not. But the gap is large enough that **rate, base and cap must all be rule-table values from the first release**: charging on the manager's fee instead of gross rent, capping the fee per payout, or tiering by monthly volume must each be a configuration change, never a rewrite. A per-payout cap is the cheapest insurance and should exist from day one even if it is set high.

RESIDUAL STREAMS

Stream	Basis	Notes
Platform fee	2.99%, once per cycle, deducted at payout	The whole business at launch
Premium AI tier	₹499 per organisation per month (future)	The only paid plan that will ever exist — see §5.6
Sale brokerage	Share of a negotiated brokerage on a completed sale	Separately contracted; not a platform fee percentage
Partner referrals	Insurance, deposit alternatives, home loans, movers, utilities	Referral fee only; dwellm8 carries no risk and discloses the relationship

5.6 The premium AI tier — ₹499 per month

Everything operational is free. The **only** paid plan is AI, priced at **₹499 per organisation per month** and introduced after the core platform is proven. This keeps the free product genuinely complete while giving the platform a second, predictable revenue line that does not depend on transaction volume.

Free forever Every module in section 6 — properties, tenancies, agreements, collections, deposits, maintenance, marketplace, hostel, commercial, accounting, documents, listings, reporting. No limits on units, beds, staff or transactions.	Premium — AI only Assistive intelligence layered on top of that data: drafting, extraction, prediction, triage, summarisation and conversational access. Removing it degrades convenience, never capability.
--	---

The entitlement rule. AI is the only gate in the product, it is enforced server-side per organisation, and no AI feature may ever become load-bearing for a core workflow. A user who never pays ₹499 must be able to run a property end to end without a single blocked action. Every AI output is advisory and attributable, and financially meaningful actions always require explicit human confirmation.

6. Module requirements and user stories

Twenty modules, each with its scope, its rules and its high-level stories. Nineteen are free to every organisation; only the AI tier of §5.6 is paid. Stories are epic-level: each will decompose into several engineering stories on the delivery board using the standard template. Priorities are as defined in §2.2.

M1 · Identity, organisations and access

Purpose. Establish who a person is, which organisation they are acting for, and what they may do — on a phone, with a phone number, on a shared low-end device.

Key rules. Phone-number-first authentication with OTP through Google Identity Platform for *every* user, including platform admins. Authorisation is a fail-closed OpenFGA check at the service boundary, with RBAC roles layered on the relationship graph — a token never carries authority. Roles and relationships are scoped to (organisation, property, unit) . A manager acting for an owner always traverses an explicit, effective-dated mandate tuple. Context switching between organisations is explicit and visible. See §9.4.

ID	As a	I want	So that	Pri
IDN-01	Any user	to sign up and sign in with my phone number and an OTP	I can start using the app without an email address or a password I will forget	P0
IDN-02	Owner	my workspace created automatically on first sign-in	I can add my first property in under two minutes with no onboarding call	P0
IDN-03	Any user	to hold several roles across different organisations under one login	I do not need three accounts to be an owner, a tenant and a guardian	P0
IDN-04	Any user	an explicit, visible context switch between my roles	I always know which hat I am wearing and never act in the wrong one	P1
IDN-05	Manager	to invite staff and assign them scoped roles over named properties	my team sees exactly what their job requires and nothing more	P1
IDN-06	Owner	to grant a manager authority over specific units and revoke it instantly	I stay in control of my asset and can end the relationship the day I choose	P0
IDN-07	Owner	to add a co-owner or family member with defined rights	my spouse can act when I am travelling without sharing my login	P2
IDN-08	Manager	an audit trail of every action taken by my staff and by the platform	I can answer an owner's question about who did what, when	P1
IDN-09	Platform Admin	time-bound, justified and audited impersonation for support	support can resolve issues without unrestricted access to customer data	P1
IDN-10	Enterprise manager	single sign-on against my corporate identity provider	dwelldm8 passes my organisation's security review	P3
IDN-12	Manager	a full web console covering everything I can do in the app, and a full app covering everything I do in the field	I work from a desk and from a scooter without either surface being a compromise	P1
IDN-11	Any user	my account to survive a change of phone number without losing history	changing my SIM does not erase my tenancy record	P2

M2 · Property, unit and inventory

Purpose. Represent a standalone house, a four-flat building, a 240-bed hostel and an office floor in one hierarchy, with the right granularity for each.

Key rules. Property → Block → Unit, extended to Unit → Room → Bed for hostels. Every inventory object carries its own availability, price basis and occupancy history. Ownership is effective-dated. Assets and fixtures are tracked because they drive maintenance liability.

ID	As a	I want	So that	Pri
INV-01	Owner	to add a property with Indian address fields and its units	I have somewhere to attach tenancies, rent and repairs	P0
INV-02	Manager	to bulk-import hundreds of units or beds from a spreadsheet with an error report	onboarding a large building takes minutes, not a week	P1
INV-03	Hostel operator	rooms and individual beds as bookable inventory with a sharing type and price	I can sell and track occupancy at the level I actually operate	P0
INV-04	Hostel operator	a live bed availability calendar with holds and confirmed allocations	I never double-allocate a bed or lose one to a stale hold	P1
INV-05	Commercial manager	to record carpet, built-up and chargeable area per suite	rent and CAM bill on the correct area basis	P1
INV-06	Owner	to record fixtures and appliances with age, warranty and serial number	maintenance liability and warranty claims resolve themselves	P2
INV-07	Manager	to attach utility identifiers — electricity consumer number, water, gas, property tax ID	bills and transfers do not require a hunt through old paperwork	P2
INV-08	Owner	parking bays, storage and terraces as allottable sub-inventory	they can be charged, allotted and disputed properly	P2
INV-09	Owner	my ownership share and any co-owners recorded with effective dates	payouts, TDS and statements split correctly between us	P1
INV-10	Manager	a single occupancy view across every unit and bed I manage	I can see vacancy and revenue at risk in one screen	P1
INV-11	Manager	photographic condition records per unit at move-in and move-out	deposit deductions are evidenced rather than argued	P1
INV-12	Owner / Manager	segment templates for residential, hostel and commercial at property creation	onboarding is fast without losing segment-specific fields	P1
INV-14	Manager	key and access-card tracking with check-out, return and overdue alerts	I know who holds the keys to every property I manage	P2
INV-13	Platform Admin	configurable property fields, document types and regional settings	state and customer variation is handled without a release	P2

M3 · Leasing, licensing and agreements

Purpose. Turn an agreed occupancy into a valid, stamped, signed and stored instrument — a lease for residential, a licence for hostel, a registered deed for commercial.

Key rules. Templates versioned per state and instrument type. Stamp duty computed from state slabs and revalidated before stamping. Any term of 12 months or more routes to the registration path and is never quietly produced as an unregistered long lease. Renewal creates a linked successor, never a mutated record.

ID	As a	I want	So that	Pri
LSE-01	Owner / Manager	to create a tenancy with parties, rent, cycle, due day, deposit and notice period	billing, reminders and settlement all follow automatically	P0
LSE-02	Owner	an agreement generated from a reviewed template for my state and property type	the paperwork is correct without a lawyer for every tenancy	P1

LSE-03	Owner	stamp duty computed and the agreement e-stamped digitally	the agreement is enforceable without a trip to a stamp vendor	P1
LSE-04	Tenant & Owner	to sign remotely with a legally recognised electronic signature and audit trail	we can complete a tenancy from two different cities	P1
LSE-05	Tenant	an alternative to Aadhaar-based signing	I can complete the tenancy without using Aadhaar if I choose not to	P2
LSE-06	System	to block any term of 12 months or more from the unregistered path	the platform never produces a defective long-term agreement	P0
LSE-07	Owner	an escalation clause that revises rent automatically on its effective date	I never under-bill for months because I forgot the anniversary	P1
LSE-08	Owner / Tenant	expiry and renewal reminders at 60, 30 and 7 days with a one-tap renewal	a tenancy never lapses quietly with the tenant still in place	P1
LSE-09	Hostel operator	a licence agreement with a house-rules annexure and guardian counter-signature	a minor or student resident is properly and lawfully documented	P1
LSE-10	Commercial manager	lock-in, rent-free fit-out, escalation schedule and exit terms as structured fields	the billing engine honours the commercial deal rather than a PDF nobody reads	P1
LSE-11	Commercial owner	the registration path with duty, fee, appointment and the registered instrument stored	a long commercial lease is completed properly through the platform	P2
LSE-12	Tenant	to open my signed agreement in the app at any time	I never have to ask my landlord for a copy of my own contract	P1
LSE-13	Owner	to record a tenancy that was signed outside the platform	I can onboard my existing tenants without re-papering them	P1
LSE-14	Manager	early termination handled with notice, lock-in and penalty rules applied	an early exit is settled by the agreement, not by negotiation	P2

M4 · Occupant onboarding, KYC and verification

Purpose. Know who is moving in, satisfy the owner's need for assurance and the state's need for verification, without building an identity honeypot.

Key rules. Aadhaar numbers are never stored, logged or sent to analytics — verification runs through authorised flows and only the result persists. Every check requires a recorded consent artefact. Screening is advisory to the requesting owner only; there is no portable score and no cross-owner blacklist.

ID	As a	I want	So that	Pri
KYC-01	Owner / Manager	to verify a prospective occupant's identity through an authorised flow	I know who I am renting to without holding their documents	P1
KYC-02	Tenant	to complete my verification from my phone in one sitting	moving in is not delayed by a week of paperwork	P1
KYC-03	Tenant	to see and consent to exactly what is being collected and why	I know what the platform holds about me and can withdraw consent	P1
KYC-04	Owner	police tenant verification tracked with the correct form or portal for my state	I meet an obligation I would otherwise forget until it is a problem	P2
KYC-05	Owner	employment or income evidence and a previous-landlord reference, with consent	I can decide on evidence rather than instinct	P2
KYC-06	Hostel operator	guardian details, college or employer verification and an emergency contact	I can discharge my duty of care for a resident in my building	P1

KYC-07	Commercial manager	entity KYC — GSTIN, CIN, PAN, authorised signatory and board resolution	I am contracting with a verified business, not an individual claiming to be one	P2
KYC-08	Manager	a digital move-in checklist with condition photos, meter readings and key handover	the starting condition is recorded and cannot be disputed later	P1
KYC-09	Tenant	to onboard additional occupants and dependants on my tenancy	my family is recorded correctly for gate access and society records	P2
KYC-10	Platform Admin	a review queue for failed or inconclusive verifications	edge cases are resolved by a human rather than blocking a move-in	P2

M5 · Billing and collections

Purpose. Turn every occupancy into correctly computed charges and then actually collect them — over UPI, autopay, NACH, cards or recorded offline.

Key rules. Multiple charge heads with different bases per vertical. Invoices are immutable once issued, gaplessly numbered, corrected only by credit note. Currency is stored in minor units; floats never touch money. Offline cash and bank transfers are first-class, because most Indian rent still moves that way. Webhooks are advisory — money is confirmed against the provider and the settlement file.

CHARGE HEADS BY VERTICAL

Vertical	Recurring heads	One-time heads
Residential	Rent · society maintenance pass-through · parking	Deposit · brokerage · move-in charges
Hostel / PG	Stay fee (by sharing type) · mess plan · laundry · utilities · Wi-Fi	Admission fee · caution deposit · room-change fee
Commercial	Rent per sq ft · CAM · electricity (metered) · DG units · water · parking bays · signage	Deposit · fit-out charges · CAM annual true-up

ID	As a	I want	So that	Pri
BIL-01	System	invoices generated automatically ahead of the due date, prorated on join and exit	every occupant knows what is owed before it is due	P0
BIL-02	Manager	multiple charge heads on one invoice with different bases and tax treatment	hostel and commercial billing works without a workaround	P0
BIL-03	Tenant	to pay by tapping a link that opens my UPI app with the amount filled in	paying rent takes seconds, not a bank transfer and a screenshot	P0
BIL-04	Tenant	to set up autopay once so rent is collected automatically each month	I never think about rent again or pay a late fee by accident	P1
BIL-05	Guardian	to pay my child's hostel fee from my own phone and receive the receipt	the payer and the occupant can be different people, as they usually are	P1
BIL-06	Owner / Manager	to record a cash or bank-transfer payment with evidence	my ledger reflects reality rather than only gateway money	P0
BIL-07	Tenant	a receipt automatically on every payment, retrievable years later	I have proof for HRA and for any dispute	P1
BIL-08	Manager	an automatic reminder ladder over WhatsApp and SMS before and after the due date	I stop having the same awkward conversation every month	P1
BIL-09	Tenant	reminders to stop the instant my payment is recorded	I am never chased for money I have already paid	P0

BIL-10	Owner	rule-driven late fees with a grace period and a cap	arrears carry a consistent charge I never have to calculate or justify	P1
BIL-11	Commercial manager	metered utility billing from readings with photo evidence	electricity and DG are billed on consumption, not estimated and disputed	P1
BIL-12	Commercial manager	CAM billed on budget and reconciled to actuals annually	the true-up is a report, not a three-month argument	P2
BIL-13	Tenant	to pay part now and the rest later, with the balance tracked	a hard month does not become a default	P2
BIL-14	Manager	bulk collection actions and a daily collection worklist	chasing 300 units is a queue, not a memory exercise	P1
BIL-15	Operations	nightly reconciliation against the aggregator settlement with drift alerts	unmatched money is found in a day, not when a landlord complains	P0
BIL-16	Tenant	a clear breakdown of every charge on my invoice	I understand what I am paying for and dispute less	P1
BIL-18	Manager	automatic bank feeds importing account transactions daily	money that arrives by NEFT or IMPS reconciles itself instead of by hand	P0
BIL-19	Manager	bank statement import and matching for accounts without a live feed	every bank a customer uses is supported, not only the integrated ones	P1
BIL-20	Manager	rent credits, concessions and negotiated adjustments applied with a reason	a rent-free month or a goodwill credit is recorded properly rather than by editing an invoice	P2
BIL-21	Manager	one-time fee types — letting fee, onboarding, admin, renewal — charged automatically	everything I bill an owner is on the platform, not in a side invoice	P2
BIL-17	Hostel operator	mess and service charges adjusted for declared leave	residents are billed for what they consumed, which is the top fee complaint	P2

M6 · Deposits and exit settlement

Purpose. Hold the largest single sum in an Indian tenancy as a visible liability, and return it on an evidenced basis rather than an argument.

Key rules. A deposit is a liability, never income. Deductions are itemised, evidenced and acknowledged. Settlement is a workflow with a tenant response window, not an owner-only screen. Where the Model Tenancy Act is adopted, the state deposit cap is enforced at agreement time.

ID	As a	I want	So that	Pri
DEP-01	Tenant & Owner	the deposit recorded with its terms and visible identically to both of us	there is one agreed record from day one	P1
DEP-02	Manager	a move-out inspection with a checklist, photos and meter readings	the exit condition is evidenced against the move-in record	P1
DEP-03	Owner	itemised deductions each with a reason, amount and evidence	a deduction is defensible rather than a round number	P1
DEP-04	Tenant	to review, accept or dispute each deduction within a defined window	I am not simply told what I am getting back	P1

DEP-05	Tenant	the undisputed portion refunded immediately while a dispute is resolved	a disagreement over ₹4,000 does not hold up ₹1.6 lakh	P1
DEP-06	System	outstanding dues, utilities and prorated rent netted in from the ledger	the settlement is complete and reconciles to the paisa	P1
DEP-07	Owner	deposit interest accrued where the state or society requires it	a statutory obligation is met without manual computation	P2
DEP-08	Tenant	a partner-provided alternative to a large cash deposit	I can move without blocking several months of income	P3
DEP-09	Commercial manager	a bank guarantee recorded as a deposit instrument with its expiry tracked	an expiring guarantee is renewed before it lapses	P3

M7 · Maintenance and ticketing

Purpose. Take a problem from "the geyser is leaking" to a completed, evidenced, paid-for repair — with the liability settled before anyone is dispatched.

Key rules. Every ticket evaluates the cost-sharing matrix at creation and shows the outcome to both parties. Every completed job produces a ledger posting against the liable party. SLAs are per category and priority, with automatic escalation. Photographic evidence at raise, before, after and sign-off.

TICKET LIFECYCLE

raise → triage & categorise → liability evaluated & disclosed → quote (if vendor)
 → approval (auto / manager / owner / tenant) → schedule → technician on site
 → execute → evidence & OTP sign-off → invoice → ledger posting → rating
 → warranty window → (rework, free, if within warranty)

ID	As a	I want	So that	Pri
MNT-01	Tenant	to raise a request in under 30 seconds with photos and a category	reporting a problem is easier than sending a WhatsApp message	P1
MNT-02	Tenant & Owner	to be told who pays, and why, before any work is approved	the argument happens once, at the rule level, not per repair	P0
MNT-03	Tenant	to track my request through every stage with expected timings	I know something is happening without having to chase	P1
MNT-04	Manager	a single queue of every open ticket across my portfolio with SLA status	nothing is forgotten and breaches are visible before they happen	P1
MNT-05	Manager	automatic escalation when an SLA is breached	a silent ticket becomes someone's problem automatically	P2
MNT-06	Tenant	to choose a convenient visit slot and be told when the technician is on the way	I do not lose a day waiting at home	P1
MNT-07	Tenant	to sign off on completion with an OTP and rate the work	a job is not closed just because the vendor says so	P1
MNT-08	Owner	a maintenance history per unit with costs, vendors and recurrence	I can see that the same tap has been fixed three times and replace it	P2
MNT-09	Commercial manager	planned preventive maintenance schedules and AMC contract tracking	lifts, DG, fire systems and HVAC are serviced before they fail an inspection	P2
MNT-10	Hostel warden	bulk and recurring tasks — housekeeping rounds, pest control, water tank cleaning	routine operations are scheduled rather than remembered	P2

MNT-11	Tenant	an emergency path for water, electrical, gas, lift or safety issues	a genuine emergency is not queued behind a squeaking door	P1
MNT-12	Manager	rework tracked against the original job under warranty at no extra cost	a badly done repair is the vendor's problem, not the owner's	P2
MNT-14	Manager	scheduled routine inspections with notice to the tenant, photos, comments and a signed report	periodic inspection is a workflow, not a diary entry and a camera roll	P1
MNT-15	Field agent	a day's inspections planned into a route with map ordering	I am not driving across the city twice in one morning	P2
MNT-16	Tenant	to complete a guided self-inspection from my phone when that is agreed	a routine check does not need someone to visit and disturb me	P3
MNT-17	Owner	an inspection report with photos and condition notes delivered automatically	I can see the state of my property without being there	P1
MNT-18	Manager	a compliance register per property with renewal dates and evidence	fire safety, lift, electrical, water tank, insurance and society NOCs never lapse unnoticed	P1
MNT-13	Owner	to approve or decline a quote from a push notification in one tap	a ₹3,000 decision does not need me to open a laptop	P1

M8 · Service provider marketplace

Purpose. Give every property access to verified plumbers, electricians, carpenters, painters, pest controllers, AC and appliance technicians and cleaners — and give those providers a steady, paid, rated stream of work.

Key rules. Providers are onboarded partners, never platform employees. Technicians are individually verified. Pricing comes from a published schedule of rates, with variance justified. Payment is released on evidenced completion. GST and §194-C TDS apply to vendor invoices. Ratings drive dispatch, and persistent poor performance de-lists.

CATEGORY COVERAGE AT LAUNCH

Tier	Categories
Core (launch)	Plumbing · Electrical · Carpentry · Appliance repair (AC, geyser, washing machine) · General handyman
Scheduled	Painting · Deep cleaning · Pest control · Water tank cleaning · Housekeeping
Specialist	Masonry & waterproofing · False ceiling · Locksmith · CCTV & intercom · RO & chimney · Lift and DG AMC
Move-related	Packers & movers · Utility connection and transfer · Home sanitisation

ID	As a	I want	So that	Pri
VND-01	Service provider	to onboard with my firm details, categories, service area and rate card	I can start receiving jobs without a sales relationship	P1
VND-02	Platform Admin	to verify a provider — KYC, GST, bank, skill evidence and technician background checks	only verified people enter customers' homes	P1
VND-03	Manager	a job auto-dispatched to the best-matched available provider	I do not personally know a plumber in every locality I manage	P1
VND-04	Technician	a mobile job list with address, contact, problem, photos and navigation	I arrive prepared and on time	P1
VND-05	Technician	to start and complete a job with an OTP from the occupant	attendance and completion are proven, not claimed	P1

VND-06	Provider	to quote from a published schedule of rates, splitting labour and materials	pricing is consistent and disputes about the bill disappear	P1
VND-07	Owner / Tenant	a second comparative quote above a threshold value	a large spend is tested against the market before I approve it	P2
VND-08	Technician	to capture before and after photos and any parts replaced	the work is evidenced and warranty claims are straightforward	P1
VND-09	Provider	to be paid on a predictable cycle with GST and TDS handled correctly	working through the platform is better than working for cash	P1
VND-10	Tenant	to rate the technician and the outcome separately	a good technician from a mediocre firm is still recognised	P2
VND-11	Platform Admin	performance monitoring with automatic suspension on repeated failures	quality is enforced by the system, not by complaints	P2
VND-12	Manager	to bring my own trusted vendors onto the platform alongside marketplace ones	I do not have to abandon relationships I have built over years	P1
VND-13	Owner	a warranty period on every job with free rework inside it	I pay once for a repair that should have worked the first time	P2
VND-14	Tenant	to book a service directly for something I am paying for myself	the app is useful to me even when my landlord is not involved	P2
VND-17	Manager	supplier bills captured by scan or upload and allocated to the right property and vendor	a stack of bills becomes ledger entries without manual data entry	P1
VND-18	Manager	recurring supplier bills and outgoings — society dues, AMC, insurance, property tax	predictable costs are scheduled rather than remembered	P2
VND-19	Owner / Manager	multi-step bill approval above a threshold	no single person can approve and pay a bill alone	P1
VND-15	Manager	AMC packages sold per property with scheduled visits	maintenance becomes predictable revenue rather than reactive cost	P3
VND-16	Provider	an earnings dashboard with jobs, ratings and settlement history	I can run my business from the app I take jobs on	P2

M9 • Cost sharing, mandate and approvals

Purpose. Encode the middle-man economics: who may commit whose money, how a cost is split, and how it is recovered — without any party having to trust another party's spreadsheet.

Key rules. The liability matrix is set on the agreement, evaluated at ticket creation, and disclosed before spend. The manager's authority is a mandate object with a per-job and monthly ceiling. Every recovery is a ledger posting, visible identically to everyone entitled to see it.

ID	As a	I want	So that	Pri
CST-01	Owner	to set a liability matrix by category, cause and value on the agreement	who pays is decided once, in advance, for every future repair	P0
CST-02	Owner	to give my manager a spend authority per job and per month	small things get fixed without me, and large things never happen without me	P0
CST-03	Manager	a higher emergency ceiling with mandatory notification	a burst pipe at midnight is fixed, not debated	P1
CST-04	Owner	to fund a maintenance float that auto-replenishes at a trigger	my manager is never blocked for want of ₹3,000	P2

CST-05	System	to split a single cost across owner, tenant and operator by percentage or threshold	a shared cost is settled by arithmetic rather than negotiation	P1
CST-06	Tenant	a tenant-borne cost added transparently to my next invoice with the evidence attached	I can see what I am being charged for and why	P1
CST-07	Owner	owner-borne costs netted from my payout with each line traceable	my statement explains itself without a phone call	P1
CST-08	Owner	a one-tap approval or decline with the quote, photos and rule shown	approving spend takes ten seconds from anywhere in the world	P1
CST-09	Manager	escalation when an owner does not respond to an approval within a window	a repair is not blocked indefinitely by an unreachable owner	P2
CST-10	Tenant	to dispute a cost allocated to me, with the manager as first arbiter	a wrong allocation has a route to correction	P2
CST-11	Owner	my manager's fee, markup and commission disclosed in full	I know exactly what the middle layer costs me	P1
CST-12	Manager	to configure fee models per owner, including per-bed pricing for hostels	my commercial terms differ by client and the system supports that	P1
CST-14	Owner / Manager	recoverable and non-recoverable expenses tracked separately with ageing	I can see what I will actually get back and what I have simply spent	P2
CST-15	Owner / Manager	to end or transfer a mandate with a handover checklist and financial reconciliation	changing manager does not lose records, deposits or continuity	P2
CST-13	Owner	a deferred tenant-borne cost carried to deposit settlement	a small recovery at exit does not need a separate collection now	P2

M10 · Accounting, payouts and tax

Purpose. Make every rupee explainable years later, settle owners accurately, and meet GST and TDS obligations as workflow rather than advice.

Key rules. Double-entry ledger, immutable postings, reversing entries with reason codes, balances always derived. Statutory rates and thresholds live in versioned, state- and date-scoped rule tables. Payouts run as durable workflows with compensations — the failure "tenant debited, owner not credited" must be structurally impossible.

ID	As a	I want	So that	Pri
FIN-01	System	every money event posted to a double-entry ledger with derived balances	no service can invent, store or drift a balance	P0
FIN-02	Accountant	corrections made only as reversing entries with a reason code and an author	the ledger remains evidence rather than a working document	P0
FIN-03	Owner	a scheduled payout net of fee, GST, TDS and shared costs	I receive my money on a predictable date without asking	P1
FIN-04	Owner	a statement in which every line traces to a posting	I trust the numbers without keeping my own spreadsheet	P1
FIN-05	Manager	my management fee computed, invoiced with GST and netted automatically	my own revenue is billed correctly without manual invoicing	P1
FIN-06	System	GST determined by place of supply, registration status and reverse-charge rules	invoices are compliant from the first one, across all three verticals	P1
FIN-07	Tenant (business)	§194-I TDS computed, posted and evidenced monthly	my company meets its deduction obligation without a spreadsheet	P2

FIN-08	Tenant (individual)	§194-IB deduction, Form 26QC and Form 16C tracked to completion	I meet an obligation I did not know I had, before it becomes a penalty	P2
FIN-09	Tenant of an NRI owner	the §195 obligation surfaced prominently before the tenancy starts	I am not exposed to a liability nobody told me about	P1
FIN-10	Owner	an annual income statement and a tenant rent-paid statement for HRA	tax time is a download rather than a reconstruction	P1
FIN-11	Provider	vendor payouts with GST and §194-C TDS handled and certificates tracked	working through the platform is tax-clean for me too	P2
FIN-12	Manager	an export in the format my chartered accountant actually uses	year end does not require re-keying twelve months of data	P2
FIN-13	Operations	payouts executed as compensating workflows with verified settlement	a partial failure can never leave money and the ledger out of step	P0
FIN-14	Owner	refunds and adjustments handled with correct postings and evidence	money returned is as traceable as money collected	P2
FIN-18	Owner / Manager	payouts triggered by a date or by an available balance threshold	owners are paid on the rhythm they chose, not when someone remembers	P2
FIN-19	Manager	bulk disbursement and a bank batch file for payments outside the platform rails	paying ninety owners is one action, not ninety transfers	P2
FIN-20	Accountant	a period close with a locked month and an audit-ready pack	the books for a closed month cannot silently change afterwards	P1
FIN-16	Platform Admin	bank and settlement details verified before collections or payouts are enabled	money never moves to an unverified account	P0
FIN-17	Manager	the platform fee computed, disclosed and posted per the organisation's bearer setting	everyone sees what the platform costs, on the invoice, before paying	P0
FIN-15	Hostel operator	revenue reporting per bed, per room type and per occupancy month	I can see yield rather than just total collections	P2

M11 · Hostel, PG and co-living operations

Purpose. Run a bed-based property day to day — allocation, mess, attendance, leave, discipline and guardian visibility — on top of the same core.

Key rules. The bed is the inventory object. The payer is frequently the guardian, not the occupant. Churn is high, so joining and vacating must be near-instant. Duty of care obligations mean attendance and emergency contact are operational requirements, not features.

ID	As a	I want	So that	Pri
HST-01	Warden	to allocate a resident to a specific bed with a joining date	occupancy is accurate at the level I actually operate	P1
HST-02	Warden	to move a resident between beds or rooms mid-term with the history preserved	a room change does not corrupt billing or occupancy records	P1
HST-03	Resident	to join mid-month and be billed proportionately	I pay for the days I actually stay	P1
HST-04	Resident	to give notice and vacate with dues and caution deposit settled in the app	leaving is as easy as joining and I get my deposit back	P1
HST-05	Warden	a mess plan with a published menu and meal opt-outs	food is planned and wastage is reduced	P2

HST-06	Resident	to declare leave and have mess charges adjusted	I am not billed for meals I did not eat	P2
HST-07	Warden	an in-out register with entry, exit and late-return flags	I know who is in the building, which is my legal duty of care	P1
HST-08	Guardian	visibility of my child's attendance, leave and fee status	I have peace of mind without calling the warden	P1
HST-09	Guardian	to approve a leave or night-out request from my phone	permissions are recorded rather than claimed over a phone call	P2
HST-10	Resident	to raise complaints about my room, mess, Wi-Fi or housekeeping	day-to-day problems have a route that is not a corridor conversation	P1
HST-11	Warden	housekeeping and laundry schedules with completion tracking	routine service is delivered and verifiable	P2
HST-12	Operator	occupancy, yield and vacancy analytics per room type and per property	I can price and fill beds with data rather than instinct	P2
HST-13	Prospective resident	to enquire, book a visit and reserve a bed with a token payment	I can secure a place before I arrive in a new city	P2
HST-14	Warden	an asset register per bed — mattress, cupboard, key, chair	missing items at vacating are recoverable and evidenced	P3
HST-15	Warden	emergency contact, medical note and an SOS from the resident app	a night-time emergency has a defined, fast path	P2
HST-16	Operator	staff attendance and shift rosters for wardens, cooks and housekeeping	the property is always staffed and payroll inputs are accurate	P3

M12 · Commercial operations

Purpose. Handle the mechanics that only commercial has — CAM, meters, fit-out, escalation schedules, compliance and high-value obligations.

Key rules. Several concurrent charge heads on different bases. CAM is budgeted and trued up. Escalations are scheduled years in advance. Registration is mandatory for long terms. GST and reverse charge apply, and the tenant is a business with procurement expectations.

ID	As a	I want	So that	Pri
COM-01	Manager	rent billed on chargeable area with a scheduled escalation ladder	a nine-year deal bills itself correctly without annual intervention	P1
COM-02	Manager	CAM billed monthly on budget and reconciled annually to actuals	the true-up is evidenced and the tenant accepts it	P2
COM-03	Manager	meter readings captured with photos and billed at published tariffs	electricity, DG and water are billed on consumption	P1
COM-04	Tenant entity	a GST-compliant invoice with PO reference and my entity details	my finance team can process it without a query	P1
COM-05	Manager	a fit-out approval workflow with drawings, contractor details and a deposit	tenant works do not damage the building or breach compliance	P2
COM-06	Manager	lock-in, notice and exit dilapidation terms enforced by the system	an early exit is settled by the deed rather than by argument	P2
COM-07	Manager	building compliance tracking — fire NOC, lift certificates, AMC renewals	an expiring certificate is renewed before an inspection finds it	P2

COM-08	Owner	parking bays, signage rights and common-area usage allotted and charged	ancillary revenue is billed rather than given away	P3
COM-09	Tenant entity	multi-site visibility across every premises I lease from this landlord	a national tenant manages a portfolio, not individual sites	P3
COM-10	Manager	tenant insurance and indemnity certificates tracked with expiry alerts	a lapsed policy is caught before an incident, not after	P3

M13 · Communication and notifications

Purpose. Reach every party on the channel they actually read, without becoming noise.

Key rules. WhatsApp is the primary channel in India; SMS is the fallback; push serves app users; the in-app inbox is the durable record. Every notification is idempotent, suppressible and rate-limited per recipient. Statutory and transactional messages are distinguished from promotional ones.

ID	As a	I want	So that	Pri
NTF-01	Tenant	rent, receipt and ticket updates on WhatsApp	I see them where I already live, not in an unread email	P1
NTF-02	Any user	an in-app inbox holding every notice ever sent to me	nothing important disappears into a chat history	P1
NTF-03	Manager	to broadcast a notice to a building, a floor or a whole portfolio	water shutdowns and lift outages reach everyone affected	P2
NTF-04	Manager	delivery and read reporting on important notices	I can prove a notice reached the people it had to reach	P2
NTF-05	Any user	quiet hours and per-category notification preferences	I am not woken by a reminder for a bill due next week	P2
NTF-06	Tenant & Manager	an in-app conversation thread attached to a ticket or a tenancy	the conversation and its context live in the same place	P2
NTF-10	Manager	a template library for email, SMS, WhatsApp and letters with merge fields	routine correspondence is consistent and takes seconds	P2
NTF-11	Manager	replies to arrive back in one inbox, whichever channel the tenant used	a reply by SMS does not vanish into someone's personal phone	P2
NTF-12	Any user	a time-stamped activity feed on every property, tenancy, ticket and contact	anyone picking up the file can see the whole history at a glance	P1
NTF-13	Manager	to mention a colleague on a record and have it reach their queue	handing over a task does not require a separate message	P3
NTF-08	Manager	surveys and feedback requests to residents or owners	I hear about a problem before it becomes a vacancy	P3
NTF-09	Resident	a community board for a building or hostel	residents help each other without a WhatsApp group nobody moderates	P3
NTF-07	Any user	the interface and messages in English or Hindi at minimum	the product works for the people who actually use it	P2

M14 · Document vault and compliance

Purpose. Hold every document a property generates over its life, retrievable years later by the parties entitled to see it.

Key rules. Documents are stored with per-organisation isolation and short-lived signed access. Retention follows a matrix that reconciles erasure rights with statutory retention of financial and executed records. Sensitive identifiers are

excluded, encrypted or never stored at all.

ID	As a	I want	So that	Pri
DOC-01	Owner	one vault per property holding agreements, receipts, bills, certificates and photos	everything about my asset is in one place I can hand over	P1
DOC-02	Tenant	permanent access to my own agreements, receipts and statements	I keep my record even after I move out	P1
DOC-03	Manager	expiry tracking on every dated document with reminders	nothing lapses unnoticed — NOCs, AMCs, insurance, guarantees	P2
DOC-04	Owner	title, tax receipts and approval documents stored for a future sale	due diligence at sale time is a folder, not an archaeology project	P2
DOC-05	Data principal	working access, correction, erasure and grievance paths	my privacy rights are exercisable rather than theoretical	P1
DOC-06	Platform Admin	a retention matrix enforced automatically	we neither hoard data nor delete records we are obliged to keep	P2

M15 · Discovery, listings, inspections and brokerage

Purpose. Extend the platform from operating a property to transacting it — helping a prospect *find* a property, *connect* with the manager and *book an inspection*; then filling the vacancy, and ultimately selling the asset — using the tenancy, maintenance and document history the platform already holds.

The discovery loop. A vacancy in the operating system becomes a public listing; a prospect finds it without an account; they enquire and are connected to the manager or owner; they book an inspection slot; the visit outcome is captured; and the accepted prospect converts into a tenancy with every field carried over. No re-keying at any step, and no listing that describes a unit which is not actually available.

Key rules. RERA and advertising rules constrain what a listing may claim. A listing is generated from real inventory, so it cannot describe a unit that does not exist or is already occupied. dwellm8 does not take custody of transaction money; token and consideration flow between the parties through regulated rails. The document vault is the platform's structural advantage at sale time.

RENTAL FUNNEL	vacancy → listing → enquiry → qualification → site visit → offer → agreement → move-in checklist → ACTIVE TENANCY (M3)
SALE FUNNEL	intent to sell → valuation & pricing → listing → lead → site visit → negotiation → token advance & agreement to sell → due diligence → loan (partner) → registration & stamp duty → handover → post-sale transfers (utilities, tax, society NOC) → brokerage settled

ID	As a	I want	So that	Pri
SAL-01	Owner / Manager	to publish a rental listing from an existing unit in a few taps	filling a vacancy does not mean re-entering the property elsewhere	P2
SAL-02	Prospective tenant	to browse verified listings with real photos, rent and availability	I am not wasting my time on listings that do not exist	P2
SAL-03	Agent	a lead pipeline with assignment, qualification and follow-up reminders	enquiries convert instead of decaying in a phone	P2
SAL-04	Prospect	to book a site visit for a slot that suits me and get directions	viewing a property does not need six phone calls	P2

SAL-05	Agent	to record site-visit feedback and interest level	the owner learns why the property is not renting	P2
SAL-06	Prospect	to make an offer and negotiate terms in the app	the deal is documented from the first number quoted	P2
SAL-07	Owner	a vacancy converted straight into a tenancy with data carried over	the funnel ends in the operating product, not in a re-key	P2
SAL-08	Owner	to list my property for sale with a price benchmark from local data	I price on evidence rather than a broker's opinion	P3
SAL-09	Buyer	a verified property pack — title, approvals, tax receipts, maintenance history	I can assess a property properly before committing	P3
SAL-10	Owner	maintenance and tenancy history presented as proof of how the asset was kept	a well-run property commands its premium at sale	P3
SAL-11	Buyer & Seller	token advance and agreement to sell handled with a documented milestone plan	the transaction has a spine both sides can see	P3
SAL-12	Buyer	a due-diligence checklist with partner legal and technical verification	nothing critical is missed before I commit my savings	P3
SAL-13	Buyer	a home-loan referral to partner lenders with document pre-fill	financing runs in parallel with the transaction, not after it	P3
SAL-14	Both parties	registration, stamp duty and the appointment tracked to completion	the legal step is managed rather than left to a runner	P3
SAL-15	Buyer	post-sale transfers — society NOC, utilities, property tax — as a checklist	the loose ends that take months are closed in weeks	P3
SAL-16	Agent	brokerage computed, invoiced and split with the platform automatically	I get paid correctly without an argument at the end	P3
SAL-19	Prospect	to search listings by city, locality, budget, configuration, furnishing, sharing type or commercial area	I can find a place that actually fits, without an account	P2
SAL-20	Prospect	a listing page with real photos, the full monthly cost broken down, deposit, amenities and availability date	I know the true cost before I travel to see it	P2
SAL-21	Manager / Owner	a listing published from live inventory, with status following occupancy automatically	a let unit disappears from search the moment it is let	P2
SAL-22	Prospect	to contact the manager from the listing without exposing either party's number	I can ask a question without inviting months of broker calls	P2
SAL-23	Prospect	to book an inspection slot from the listing and get directions and a reminder	seeing a property takes two taps rather than six phone calls	P2
SAL-24	Manager / Agent	an inspection calendar with slot capacity, staff assignment and travel grouping	a day of viewings is planned rather than improvised	P2
SAL-25	Prospect & Manager	reschedule, cancel and no-show handling with automatic reminders	nobody waits at an empty flat for someone who is not coming	P2
SAL-26	Agent	to capture the inspection outcome — interest level, objections, offer, next action	the owner learns why the property is not renting	P2
SAL-27	Prospect	to shortlist listings, save a search and be alerted when a match appears	I do not have to check the app every day	P3

SAL-28	Prospect	to apply from the listing, with my details reused into KYC and the agreement	applying does not mean filling the same form three times	P2
SAL-29	Hostel operator	a bed-level listing with sharing type, gender policy, mess inclusion and available-from date	a student can find and reserve a bed before arriving in the city	P2
SAL-30	Commercial manager	a commercial listing with chargeable area, floor, fit-out status, CAM and parking	a business can evaluate the space on the terms it actually cares about	P3
SAL-31	Prospect	a shareable listing link that previews correctly on WhatsApp	I can send it to my family and decide together	P2
SAL-33	Owner / Manager	my listing published to the major Indian portals from one place	I reach the whole market without maintaining four listings	P3
SAL-32	Anonymous visitor	to browse listings and understand what dwellm8 does without creating an account	I can judge whether it is worth signing up before I hand over my number	P2
SAL-17	Platform Admin	listing moderation and compliance guardrails on claims	the platform is not liable for what an agent wrote at midnight	P2
SAL-18	Seller	outstanding society dues and tenancy encumbrances flagged before listing	a sale does not collapse at the NOC stage	P3

M16 • Platform administration

Purpose. Let dwellm8's own team operate the platform — onboard, configure, support, reconcile and intervene — without engineers running queries in production.

ID	As a	I want	So that	Pri
ADM-01	Platform Admin	organisation, user, tenancy, invoice and payment lookup with full audit	support resolves issues without engineering or database access	P1
ADM-02	Platform Admin	approval queues for manager, vendor and agent onboarding	only verified operators reach customers	P1
ADM-03	Platform Admin	to maintain statutory rule tables — GST, TDS, stamp duty, deposit caps, rates	a Budget change is a data update, not a release	P1
ADM-04	Platform Admin	platform-fee configuration, bearer settings and AI entitlement per organisation	commercial terms change without an engineering ticket	P1
ADM-10	Platform Admin	instrument-level margin monitoring — fee earned against aggregator cost	a payment mix that loses money is caught in days, not at quarter end	P1
ADM-05	Platform Admin	a reconciliation and dispute desk with ageing and ownership	unmatched money and open disputes have a queue and an owner	P1
ADM-06	Platform Admin	in-flight workflow inspection and safe operator actions on stuck operations	a stuck payout is resolved without a database edit	P2
ADM-07	Platform Admin	redaction rules so support never sees KYC identifiers or bank details	supporting a customer does not mean seeing everything about them	P1
ADM-08	Platform Admin	platform health — collections, dispatch, delivery, drift — in one dashboard	we know something is broken before a customer tells us	P1
ADM-11	Platform Admin	push alerts for platform health, SLO breach and reconciliation spikes, with acknowledgement	on-call learns about an incident without watching a dashboard	P2
ADM-12	Platform Admin	to clear approval queues and triage disputes from my phone with the same permissions and audit trail	a verification or a stuck payout does not wait until I reach a desk	P2

ADM-13	Platform Admin	administrative surfaces reachable only by an authenticated identity with MFA, with no anonymous or demo path	the control plane cannot be reached by anyone who has not proven who they are	P0
ADM-09	Platform Admin	content moderation and takedown for listings and uploaded media	the marketplace stays lawful and usable	P2

M17 · Analytics and reporting

ID	As a	I want	So that	Pri
RPT-01	Owner	a home screen answering "did I get paid this month" in five seconds	the most common question needs no navigation	P1
RPT-02	Manager	collection efficiency, arrears ageing, occupancy and SLA breach across the book	I manage by exception rather than by memory	P1
RPT-03	Owner	yield, cost and vacancy per unit over time	I can see which property is actually worth keeping	P2
RPT-04	Manager	a maintenance spend analysis by category, vendor and property	I can renegotiate rates and spot a recurring failure	P2
RPT-05	Operator	hostel occupancy and revenue per bed by room type and season	pricing and capacity decisions are evidence-based	P2
RPT-06	Manager	scheduled reports delivered to owners automatically	reporting is not a monthly manual ritual	P2
RPT-11	Manager	portfolio KPIs — properties gained and lost, response times, overdue tasks, inspection compliance	I can see whether the business is improving, not just what is outstanding today	P2
RPT-12	Manager	to switch between a property view and a client view of the same data	I can work the way the question is asked — by building or by person	P2
RPT-08	New owner or manager	a populated demonstration workspace on first sign-in, clearly marked as sample data	I can see exactly what the product does before entering anything of my own	P1
RPT-09	New owner or manager	to explore the sample portfolio — open a tenancy, raise a ticket, walk an approval, view a statement	I learn the product by using it rather than by reading a tour	P2
RPT-10	New owner or manager	the demonstration workspace removed entirely the moment I add my first real property	sample figures can never be confused with my money	P1
RPT-07	Platform Admin	anonymised market benchmarks above a minimum aggregation threshold	customers get insight without any organisation being identifiable	P3

M18 · AI capabilities *(premium tier — ₹499 per organisation per month)*

Purpose. Layer assistive intelligence over data the platform already holds. This is the only paid part of the product, and it is deliberately additive.

Key rules. AI is **never load-bearing** — a non-paying organisation can run a property end to end without a blocked action. Every output is advisory, attributable and reviewable. Financially meaningful actions always require explicit human confirmation; no AI action moves money, signs a document or dispatches a vendor on its own. Customer data is not used to train models, and entitlement is enforced server-side.

ID	As a	I want	So that	Pri
----	------	--------	---------	-----

AIX-01	Manager	a maintenance request auto-triaged from its photo and description into category, severity, likely liability and suggested vendor	a hundred tickets a day are sorted before I open the queue	P2
AIX-02	Manager	a vendor quote checked against the schedule of rates and past jobs, with outliers flagged	I spot an inflated quote without knowing every market rate	P2
AIX-03	Manager	arrears risk predicted per tenancy from payment history and behaviour	I chase the accounts that will actually default, not all of them equally	P2
AIX-04	Manager / Owner	an uploaded agreement, bill or statement read and its fields extracted automatically	migrating a legacy portfolio is not months of typing	P2
AIX-05	Owner / Manager	to ask my portfolio questions in plain language — "what is outstanding in Whitefield"	I get an answer without building a report	P3
AIX-06	Manager	notices, reminders and letters drafted in English or Hindi from context	routine writing stops consuming my day	P3
AIX-07	Manager	move-in and move-out photos compared with suggested, evidenced deductions	settlement is drafted in minutes and still defensible	P3
AIX-08	Owner	a rent or fee recommendation from comparable local data with the reasoning shown	I price on evidence rather than on what my neighbour claims	P3
AIX-09	Agent / Owner	listing copy generated from real unit data and photos, within advertising rules	a vacancy is published in minutes without inventing claims	P3
AIX-10	Manager	anomalies flagged — unusual utility consumption, duplicate billing, unexpected ledger movement	a leak or a mistake is found before it becomes a quarter's loss	P3
AIX-11	Support / Manager	a tenancy or dispute history summarised before a conversation	I walk into a call already knowing the facts	P3
AIX-12	Operator	occupancy, churn and vacancy forecasts per property or room type	I plan capacity and pricing ahead of the season	P3
AIX-16	Manager	state-specific compliance guidance in context, citing the rule	I get an answer where the work is, instead of searching a government site	P3
AIX-13	Owner / Manager	a free trial, then to subscribe or cancel the AI tier in the app in one step	I can judge whether ₹499 is worth it on my own data	P2
AIX-14	Platform Admin	AI entitlement enforced server-side with per-organisation usage and cost attribution	the tier is neither bypassable nor unprofitable	P2
AIX-15	Any user	every AI output labelled as assistive, with its source data and a correction path	I know what the machine suggested and what a person decided	P2

M19 · Demo, sandbox and trial experience

Purpose. One sandbox, two jobs. **Before sign-up:** let anyone — a landlord, a management firm, a hostel operator, a prospective tenant — use the real product with no account at all, then continue into a real account without starting again. **After sign-up:** give existing customers and their staff a safe place to **test and train on new features and releases** without touching live money, live tenants or live vendors.

The second job is the one that keeps the sandbox honest. A demo built purely for marketing rots, because nobody depends on it. A sandbox that a manager uses to train a new field agent every month, and that customers use to try each release, stays current because it is part of how the product ships.

Key rules. No sign-up, no phone number and no email is required to start a demo — that is what makes it usable both by a stranger evaluating the product and by an existing customer's new staff member on their first day. **Every demo or**

sandbox account is fully isolated: its own ephemeral organisation, seeded from a template, private to that session or that user, enforced by the same tenancy and authorisation mechanisms that separate two paying customers — never a shared mutable account that people edit over each other, and never a flag on a real organisation. Isolation is what makes the sandbox safe to leave open to the internet and safe to hand to a trainee. Everything in §9.6 applies without exception: marked as sample, isolated, structurally inert, non-routable contacts, excluded from all real reporting. **There is no administrative demo.** The demo uses the real product's components and flows, not a mock — what is demonstrated must be what is delivered.

ID	As a	I want	So that	Pri
DM0-01	Visitor	to start a demo in one tap from the website or the app, with no sign-up, phone number or email	I can try the product before deciding to give anyone my details	P1
DM0-02	Visitor	to choose the demo that matches me — landlord, management firm, hostel operator, commercial, or tenant	I see my own situation, not a generic showcase	P1
DM0-03	Visitor	my own private sandbox rather than a shared account other people are editing	what I change stays mine and nobody else's mess confuses me	P1
DM0-04	Visitor	to actually operate the sandbox — record a payment, raise a ticket, approve a quote, settle a deposit, generate a statement	I judge the product by using it, not by watching a tour	P1
DM0-05	Visitor	to switch between roles on the same sample property — see it as the owner, then the manager, then the tenant	I understand how the parties actually connect, which is the whole point of the platform	P2
DM0-06	Visitor	my demo session to survive closing the browser or the app for a few days	I can come back and continue instead of starting over	P2
DM0-07	Visitor	to reset the sandbox to its starting state at any time	I can experiment freely without fear of ruining the demo	P2
DM0-08	Visitor	to continue into a real account from inside the demo, keeping my settings and my place	signing up is a continuation, not a restart	P1
DM0-09	New customer	the sandbox retired completely the moment I add my first real property	sample figures can never sit beside my actual money	P1
DM0-10	Visitor on mobile	the same demo inside the apps, without signing in	I can evaluate the app my tenants will actually use	P2
DM0-11	Sales or partner	a pre-seeded demo for a specific vertical that I can walk a customer through live	a demonstration is consistent and never depends on the wifi in someone's office	P2
DM0-12	Product team	demo funnel analytics — entry, flows completed, drop-off, conversion to a real account	we learn which parts of the product actually sell it	P2
DM0-13	Platform	demo sessions rate-limited, expiring and cost-bounded	a free, unauthenticated sandbox cannot be turned into an abuse or cost problem	P1
DM0-14	Platform	every demo session provably inert — no posting, payment, fee, message, webhook or dispatch	the feature that sells the product can never damage it	P0
DM0-15	Visitor	never to be shown an administrative surface in any demo	the control plane is not advertised, demonstrated or reachable	P0

DMO-16	Existing customer	a sandbox I can open from inside my real account, seeded to look like my own portfolio	I can try something risky without doing it to my actual tenants and money	P2
DMO-17	Manager	to train new staff in the sandbox before giving them access to the real book	a new field agent learns on sample data instead of on a real arrears call	P2
DMO-18	Existing customer	to try a new release in the sandbox before it reaches my live account	a change to how I get paid never surprises me on the first of the month	P2
DMO-19	Any user	a what's-new walkthrough that runs on sample data rather than my own	I can learn a new feature by trying it, not by reading release notes	P2
DMO-20	Manager or platform	training completion visible per staff member	I know who has actually been trained before they touch a live tenancy	P3

M20 · Workflow automation

Purpose. Let a manager encode their own operating procedure — triggers, conditions and actions — without an engineer. This is what separates software a firm uses from software a firm runs on.

Key rules. Automations are configuration, versioned and auditable. Nothing automated may move money, sign a document or dispatch a vendor without the approval the mandate requires. Every automated action is attributed to the automation that caused it, and any automation can be paused instantly.

ID	As a	I want	So that	Pri
AUT-01	Manager	ready-made automations for the routine work — arrears follow-up, lease expiry, inspection scheduling, compliance renewals	the obvious things run themselves from day one	P1
AUT-02	Manager	to build my own rules from triggers, conditions and actions without code	the platform fits how my firm actually works	P2
AUT-03	Manager	one action to fire a whole checklist — a move-in, a move-out, an owner onboarding	a multi-step process is one click and nothing is skipped	P2
AUT-04	Manager	to see what an automation did, and pause it instantly	an automation that misbehaves is stopped in seconds, and its effects are traceable	P1
AUT-05	Manager	automations blocked from moving money or dispatching without the required approval	convenience never becomes an unauthorised spend	P0

7. Mobile-first experience requirements

dwelIm8 is used standing in a corridor, at a gate, on a scooter and on a ₹9,000 Android phone with two bars of signal. Mobile is not a companion to the web product; for tenants, residents, guardians, technicians and field agents it is the *only* product.

7.1 Five apps, one system

App	Primary users	Core jobs
dwelIm8 Live	Tenant, resident, guardian	Pay, receipts, raise and track a repair, agreement, deposit, notices, book a service
dwelIm8 Own	Owner	Portfolio and collections at a glance, approve spend, payouts and statements, documents, vacancy
dwelIm8 Ops	Manager, field agent, warden	Collection worklist, ticket queue, inspections, bed allocation, gate, offline capture
dwelIm8 Pro	Technician, service provider	Job list, navigation, OTP start and finish, photo evidence, quotes, earnings
dwelIm8 Admin	Platform Admin, support, on-call	Alerts and platform health, approval queues, dispute and reconciliation triage, customer lookup, on-call intervention — the mobile half of the Admin surface (§7.4)

The apps are for what happens away from a desk. Everything that needs a keyboard, a wide table or a configuration screen lives on the web — see §7.4.

7.2 Non-negotiable mobile principles

Three taps to money

From opening the app to completing a payment must be three taps or fewer. From a reminder link, one tap.

Works at one bar

Critical reads — dues, receipts, agreement, job list — are cached and viewable offline. Writes queue and sync, and never falsely report success.

Low-end first

Performance budget set against an entry-level Android, not a flagship. App size, cold start and list scrolling are measured on the target device.

WhatsApp is the front door

Most sessions start from a WhatsApp message. Deep links must land exactly on the right screen, authenticated, without a home-screen detour.

Evidence by camera

Photos are a primary input — condition, meters, jobs, receipts. Capture must be fast, compressed and resumable on a bad connection.

One-tap decisions

Approvals, sign-offs and confirmations are answerable from a notification without opening a laptop or reading a PDF.

Money is unambiguous

Amounts in the Indian numbering convention, dates in Indian format, and no screen that shows a number a user cannot trace.

Language and literacy

English and Hindi at minimum, with plain wording, iconography and no jargon in any tenant-facing string.

7.3 Journeys that must be exemplary

1. **Pay rent** — WhatsApp reminder → tap → UPI app → back → receipt. Under 30 seconds, no login friction.

2. **Report a leak** — open → photo → category → submit → immediate confirmation of who pays and what happens next.
3. **Approve a repair** — push notification with quote and photos → approve → done. No app navigation required.
4. **Move a resident into a hostel bed** — allocate → agreement → guardian signs → first invoice → all within one visit.
5. **Complete a job** — technician arrives → OTP start → work → photos → OTP finish → invoice raised automatically.
6. **Settle a deposit** — inspection photos → itemised deductions → tenant reviews and accepts → refund initiated same day.

7.4 Web surfaces — and the Admin surface across both

The property manager is the exception that proves the rule: they need both, fully. A manager runs listings, collections, dispatch and accounting from a desk, and inspections, cash collection and site visits from a phone. dwellm8 Ops is therefore delivered as a **complete web console and a complete mobile app** — peers sharing one permission model and one audit trail, not a product and its companion. Beyond that, mobile-first is a rule about the field, not a ban on the desktop. Four kinds of work are genuinely bad on a phone and must be delivered as responsive web: **configuration, wide tabular review, bulk operations and investigation across many records.** Much of an administrator's day is exactly that work — so Admin is **web and mobile together**, split by the nature of the task rather than by user.

Admin task	Surface	Why
Alerts, incidents, platform health	App first	An on-call administrator must learn about a collection outage or a reconciliation spike wherever they are, and acknowledge it immediately
Approval queues — managers, vendors, agents, KYC exceptions	App and web	A queue item is a single decision with evidence attached; it should not wait for someone to reach a desk
Customer lookup and support context	App and web	Support answers questions on the move; the same redaction and access-audit rules apply on both
Dispute and reconciliation triage	App triage, web resolution	Assigning, prioritising and escalating works on a phone; investigating and resolving needs the wide view
Workflow intervention on a stuck operation	App for safe actions, web for the rest	Retrying or escalating is one tap; anything with financial consequence stays on web with confirmation
Rule tables, fee and entitlement configuration	Web only	Statutory rates, bearer settings and thresholds are high-consequence configuration; they need review, comparison and a considered commit
Bulk operations and data corrections	Web only	Volume work with preview and error reporting
Reporting and cross-record investigation	Web only	Multi-column tables and exports

Web surface	Users	Priority	Why it cannot be a phone screen
dwellm8 Admin Console <i>web + mobile app</i>	Platform Admin, support, finance, compliance	P1 from first paying customer	Web carries the depth — rule tables, fee configuration, bulk operations, reconciliation and dispute desks, cross-record investigation, moderation. The app carries the urgency — alerts, approvals, triage and on-call intervention. One permission model, one audit trail, two form factors.

dwelIm8 Ops Web <i>full console — web and app are peers</i>	Manager, accountant, warden, agent	P1	The manager's primary workplace and a complete surface, not a back-office annexe: portfolios and tenancies, collections worklist, listings and the lead-to-lease pipeline , inspection calendar, ticket queue and dispatch, vendor management, owner statements and payouts, accounting and tax, fee and mandate configuration, bulk import, staff administration and reporting. Anything the Ops app can do, the web console can do.
Owner web <i>responsive, app-first</i>	Owner, NRI owner	P2	Annual statements, tax exports, multi-property comparison, document downloads at tax time.
Vendor firm web <i>responsive</i>	Service provider back-office	P2	Technician roster, rate cards, invoices and settlement history — the firm's admin, distinct from the technician's job app.
Warden console <i>web + tablet</i>	Warden, hostel operator	P2	Bed allocation grids, mess planning, attendance registers — grid work that needs screen area.

ADMIN — FIRST-RELEASE SCOPE

Web (P1, from the first paying customer): organisation, user and tenancy lookup with full access auditing and field redaction; onboarding approval queues for managers, vendors and agents; statutory and configuration rule tables (GST, TDS, stamp duty, deposit caps, schedules of rates, SLA targets, platform fee and bearer settings); the reconciliation and dispute desk with ageing; workflow inspection with safe operator actions; entitlement and feature control; moderation; and the platform health dashboard. Without it, every support question becomes an engineering task and every rate change becomes a release.

App (P2, shortly after): push alerts on platform health and SLO breach with acknowledgement; the approval queues with evidence inline; customer lookup with the same redaction rules; dispute and reconciliation triage; and a constrained set of safe on-call actions. The app is a *view onto the same system* — identical permissions, identical audit trail, no capability that exists only there. High-consequence configuration is deliberately absent from it.

8. Critical end-to-end workflows

Five workflows carry the product. Each crosses several modules and at least two personas, and each is where a partial implementation shows itself immediately.

8.1 Property and portfolio onboarding

1. Manager or owner creates the organisation and picks the **segment template** — residential, hostel or commercial — which sets default fields, charge heads and the liability matrix.
2. Property structure created or imported: blocks, units, rooms, beds, parking, meters, assets and documents.
3. Owner and manager agree the **management mandate** — scope, fee model, spend authority, float, payout schedule.
4. Bank and settlement details captured and **verified before collections or payouts are enabled**.
5. Existing occupants, balances, deposits, agreements and open maintenance items migrated, with an opening-balance reconciliation.
6. Go-live checklist reviewed and the property activated, with the whole sequence auditable.

8.2 The monthly rent and dues cycle

1. Charges generated from every active occupancy, prorated for joiners and leavers, across all applicable heads.
2. Invoices issued with an itemised breakdown; the reminder ladder starts ahead of the due date.
3. Collection over UPI, autopay mandate, NACH or card; offline cash and transfers recorded with evidence.
4. Payments allocated to invoices by deterministic rules; exceptions queued rather than guessed.
5. Platform fee computed on the transacted amount and posted (\$5.5); late fees and payment-plan rules applied only where configuration permits.
6. Nightly reconciliation against the settlement file; drift aged and alerted.
7. Owner statement, manager fee, GST, TDS and shared-cost recoveries computed; payout executed as a compensating workflow.

8.3 The maintenance and shared-cost workflow

1. Occupant reports the issue with photos, category, location and access instructions.
2. System classifies it — emergency, warranty-covered, owner-borne, tenant-borne or shared — and **discloses the outcome to both parties before any spend**.
3. Manager assigns internal staff or requests quotes from verified providers; a second quote is required above a threshold.
4. Approval routed by the mandate ladder — auto, manager, owner or tenant.
5. Technician attends, starts on an occupant OTP, completes with before/after evidence, closes on a second OTP.
6. Provider invoices; GST and §194-C TDS applied; platform commission taken.
7. Cost allocated per the split rule and posted — recovered from the payout, added to the next invoice, drawn from the float, or deferred to deposit settlement.
8. Provider settlement and both party ledgers reconcile automatically. The warranty window opens; rework inside it is free.

8.4 Move-in to move-out

1. Enquiry or allocation → screening and KYC with consent → agreement generated, stamped and e-signed.
2. Move-in inspection: condition photos, meter readings, asset checklist, key and access-card handover.
3. Occupancy runs — billing, maintenance, notices, renewals and escalations.
4. Notice served; lock-in, notice-period and penalty rules evaluated automatically.
5. Move-out inspection compared against the move-in record; itemised deductions with evidence.

6. Occupant reviews, accepts or disputes; the undisputed portion is refunded immediately; final statement issued and the ledger closed to zero.

8.5 Manager change or mandate handover

1. Owner or manager initiates termination or transfer of the mandate, with a notice period.
2. Handover checklist generated: open tickets, in-flight jobs, deposits held, float balance, outstanding fees, documents, keys.
3. Financial reconciliation to an agreed cut-off date.
4. Access revoked at the cut-off; history and documents retained and still attributable.
5. New manager onboarded onto the same property with no data loss and no tenant disruption.

9. Integrations, data principles and technology stack

9.1 Integration framework

Every external dependency sits behind an adapter with a canonical internal model. The platform must never depend on a single provider in any category, and swapping one must not touch domain code.

Category	Purpose	Requirement
Payment aggregator	Collections, mandates, payouts, refunds	Adapter with a canonical payment model; idempotency; settlement-file ingestion; a second provider addable without domain change. Aggregator cost is a direct input to platform-fee margin (§5.5)
Banking and payouts	Owner, vendor and refund settlement	Account verification before enablement; payout state verified against the provider, never webhook-only
Messaging	WhatsApp Business, SMS, push, email	Template inventory and approval; delivery receipts; opt-in state; a fallback chain proven before launch
e-Stamp and eSign	Agreement execution	State-wise channel coverage; licensed ESP; artefacts and audit trail stored; durable workflow with compensation
KYC and verification	Identity, PAN, entity, bank	Authorised flows only; result-only retention; a consent artefact per check
Accounting	Tally, Zoho Books, QuickBooks export	Export in the format the customer's chartered accountant uses; never force a migration off their existing books
Maps and routing	Addresses, technician routing, site visits	Address normalisation for Indian locality formats; offline-tolerant on the technician app
Storage and media	Documents, photos, inspection evidence	Per-organisation isolation; short-lived signed access; compression at capture
Utilities and civic	Electricity, water, property tax, DigiLocker	Opportunistic; never a hard dependency of a billing cycle
AI and model providers	Premium AI capabilities (§5.6, M18)	Provider-abstracted; per-organisation entitlement enforced server-side; no customer data used for training; cost per organisation attributable
Partner services	Insurance, deposit alternatives, home loans, movers	Referral integrations with disclosure; dwellm8 carries no risk
Public API and webhooks	Enterprise customers and partners	Versioned, organisation-scoped, rate-limited and auditable

9.2 Data model principles

- **One API, one core, extended per segment.** A single Go monolith serves every client; verticals add modules inside it, never a parallel service. Residential, hostel and commercial share organisation, property, unit, occupancy and ledger. A vertical adds inventory depth, charge heads and modules — never a parallel data model.
- **Everything effective-dated.** Ownership, rent, occupancy, membership, mandates and fee agreements are validity intervals, not overwritten rows.
- **Money is postings.** Balances are derived; there is no stored, mutable amount-due column anywhere in the system.
- **One identity, many scoped roles**, always bound to (organisation, property, unit).
- **Tenancy is enforced by the database**, not by application discipline.

- **Configuration over constants.** Rates, thresholds, SLAs, liability matrices, charge heads, entitlements and templates are versioned data with an owner and a review date.
- **Evidence attaches to the object it justifies** — the photo to the inspection line, the certificate to the agreement, the challan to the deduction.
- **Schema lives in the platform infrastructure repository;** application repositories hold ORM models only.

9.3 Technology stack

Settled, and aligned with the platform patterns already proven in the org's other products (`mark8ly` , `Home-Chef-App`) so that tooling, CI and the design system are shared rather than reinvented.

ONE API, ONE DEPLOYABLE

dwellm8 is a modular monolith, not a set of microservices. One Go API serves every client — Live, Own, Ops, Pro and Admin on mobile; the Ops, Owner, Admin and vendor consoles on web; and the public listing site. There is one OpenAPI specification, one authentication and authorisation path, one database and one transaction boundary.

This is the right call for this product, and deliberately so. The domain is *densely* interconnected — a maintenance job touches the ticket, the liability matrix, the vendor, the ledger, the owner statement and the notification ladder in a single logical operation. Splitting that across services would replace transactions with distributed sagas and turn a correctness problem into a coordination problem, for a team that does not need to scale services independently. Modularity is enforced **inside** the codebase — clear module boundaries, no cross-module database access, an internal event bus — so that if a module ever genuinely needs to be extracted, the seam already exists.

Layer	Choice	Notes
Web	Next.js with the org design system	Public listing site (SEO and server rendering matter), owner and manager consoles, and the Admin console
Backend	One Go monolith API with Gin	A single deployable API serves every client — five apps, three web consoles and the public site. Modular inside (property, lease, money, maintenance, marketplace, documents, notifications, identity), one deployment, one database, one transaction boundary.
Mobile	React Native on Expo , expo-router, NativeWind + Tailwind, @tesseract/native and design tokens	Matches the existing HomeChef and mark8ly apps exactly — same navigation, styling, icon and font conventions
Mobile data layer	TanStack Query, zustand, react-hook-form + zod, axios	Query persistence and a netinfo-aware queue give the offline tolerance §7.2 requires
Mobile platform services	expo-secure-store, expo-notifications, expo-image-picker/manipulator, expo-location, react-native-maps, expo-updates	Evidence capture, push, deep links, routing for technicians and site visits, OTA fixes
Payments in app	react-native-razorpay with UPI intent handling	Same integration already running in the org's consumer apps
Mobile build	EAS Build with a manual-dispatch CI workflow, per-app <code>eas.json</code>	Native binaries, not container images — a separate path from the web and API pipelines
Shared mobile package	A single <code>mobile-shared</code> workspace package — api client, auth, hooks, types, theme, ui, shared screens	Five apps share one implementation; a fix lands once

Identity	Google Identity Platform (Firebase Auth) for every user — consumers, staff and admin alike	Phone OTP as the primary factor, matching the existing org apps; one provider, one integration, no second identity stack to operate
Authorisation	OpenFGA for relationship-based access, with RBAC roles on top	Fail-closed decisions at the service boundary; PostgreSQL row-level security stays as the second line of defence, not a replacement for it
Data & runtime	PostgreSQL with row-level security, Redis, NATS JetStream, Temporal, GKE + Istio + ArgoCD, OpenTelemetry	Per §9.2 and §10

9.4 Identity and authorisation — GIP everywhere, OpenFGA for permissions

DECIDED

Authentication is Google Identity Platform for every user — tenant, resident, guardian, prospect, owner, manager, field agent, warden, technician, agent and platform admin. There is no second identity provider and no Keycloak.

Authorisation is OpenFGA, with conventional RBAC roles layered on top of its relationship model. Authentication proves *who you are*, exactly once; everything about *what you may do* is an authorisation decision made against the relationship graph.

The separation matters because dwellm8's permission model is inherently relational, not hierarchical. "This manager may approve spend on this unit because the owner granted a mandate scoped to it, which is still in force" is a relationship question, and expressing it as a role string in a token is how permission models rot.

WHAT THIS GIVES US

- **One identity integration.** Phone OTP is the primary factor for everyone, matching the apps already running in the org. Nothing to operate twice, and no user who exists in two providers.
- **Relationships as first-class permissions.** Ownership, delegation mandates, staff assignment, committee membership, guardian-of, technician-assigned-to-job and agent-assigned-to-listing are all tuples in the same graph.
- **Effective-dated authority genuinely enforced.** A revoked mandate or an expired committee term stops granting access because the tuple is gone, not because a nightly job remembered to strip a role.
- **Auditable decisions.** Every check can be explained — which relationship granted it — which is exactly what an owner asking "how did my manager approve that" needs.

THE RULES THIS IMPOSES

Rule	Detail
Admin is not a super-user	Platform staff authenticate through the same provider as everyone else and hold explicitly modelled administrative relationships. There is no implicit bypass, and elevated access is time-bound, justified and audited.
Tokens carry identity, not authority	A GIP token establishes the subject. It never carries a role list that a service trusts. Every authorisation decision is a check against the graph.
OpenFGA does not replace row-level security	The graph decides <i>whether</i> an action is permitted; the database still enforces tenant isolation on every query. Two independent mechanisms, and a bug in one must not be sufficient to leak data.
Tuples are derived from domain events	Ownership, mandate, staff assignment and occupancy changes write tuples transactionally with the domain change — never by a reconciliation job that can drift.
Fail closed, always	An unavailable or ambiguous authorisation decision denies. A degraded authorisation service degrades access, never expands it.
Checks are on the critical path	Every request incurs at least one check, so latency budget, caching strategy and cache invalidation on revocation are design requirements, not tuning work.

Anonymous access is modelled too	A prospect browsing listings has no identity. Public read access is an explicit relation in the model, not an unauthenticated code path that skips the check.
----------------------------------	---

9.5 Access tiers — from anonymous browsing to administrative login

Five tiers, each an explicit relation in the authorisation model. Nothing is reachable by "not being checked"; the anonymous tier is a modelled permission like any other.

Tier	Who	Can see and do	Cannot, under any circumstance
Anonymous	Prospective tenant, hostel seeker, commercial occupier, curious owner — no account	Browse and search published listings; open a listing with photos, amenities and the full cost breakdown; see approximate location; view the public product pages that explain how dwellm8 works; run a full product demo in a private sandbox (M19) ; shortlist locally on the device; share a listing	See exact addresses, occupant details, owner or manager phone numbers, any unpublished unit, any financial record, or any authenticated screen
Verified prospect	Someone who has verified a phone number to act	Enquire, message the manager through masking, book an inspection, save a search, persist a shortlist, apply for a tenancy	See anything about a property beyond what its listing publishes, or contact anyone outside the enquiry they created
Customer	Tenant, resident, guardian, owner	Everything scoped to their own occupancy, property or portfolio	Reach another organisation's data by any route
Operator	Manager, field agent, warden, technician, agent	What their assignment or mandate grants, and only for its validity period	Act beyond the mandate's scope, spend authority or dates
Administrator	dwellm8 platform staff	Explicitly modelled administrative relations, on web and the Admin app	Exist anonymously at all — see below

ADMINISTRATIVE ACCESS — AUTHENTICATED ALWAYS, NO EXCEPTIONS

There is **no anonymous or unauthenticated path into any administrative surface**, on web or mobile. No public admin route, no demo admin, no shared credential, no read-only preview. Every administrative session is an authenticated GIP identity with mandatory multi-factor authentication, a shorter session lifetime, re-authentication before sensitive actions, and a full access audit that records what was viewed as well as what was changed. The Admin console is excluded from search indexing and should be reachable only from an approved network path. If an administrative screen can be rendered without a verified identity, that is a defect of the highest severity, not a convenience.

9.6 Demonstration data — showing the product before there is anything in it

An owner who signs up has an empty dashboard, and an empty dashboard sells nothing. The first-run experience therefore presents a **fully populated demonstration workspace** — a sample portfolio with properties, tenancies, collections, arrears, tickets, vendor jobs and statements — so a prospective customer can see exactly what the product does with their own hands before entering a single real record. The same populated views drive the public product pages and sales demonstrations.

The same machinery serves a second, larger job: a **demo anyone can use before signing up at all** — see module M19. This is a marketing surface pointed at a financial product, so the rules protecting it are strict, and they apply identically to both.

Rule	Detail
------	--------

Unmistakably labelled	Persistent, non-dismissable sample-data marking on every screen and every export while demonstration mode is active. A screenshot taken from it must be self-evidently a sample.
Never mixed with real data	Demonstration records live in an isolated demo organisation. They are never queryable, joinable or aggregable with real records, and they never appear in any real balance, statement, report or analytics extract.
Inert by construction	No ledger posting, no invoice number consumed, no payment, no payout, no platform fee, no WhatsApp or SMS message, no push notification, no webhook and no vendor dispatch can originate from demonstration data. The isolation is enforced in the services, not by hoping nobody presses the button.
No real people	Sample tenants, owners, vendors and technicians are clearly fictional, with reserved phone numbers and non-routable contact details. No real person's name, photo, phone number or property ever appears.
Exits cleanly	Adding the first real property, or one explicit action, removes the demonstration workspace entirely and irreversibly. It is never merged, never partially retained, and never leaves an orphaned record behind.
Explorable, not just decorative	The user can open a tenancy, raise a sample ticket, walk an approval and view a statement. A static screenshot is not the requirement; a working sandbox is.
Never for administrators	Demonstration mode exists for owners, managers and operators evaluating the product. Administrative surfaces have no demo mode and no sample tenant — they show real platform state to an authenticated administrator, or nothing.

9.7 The authorisation model — shape, isolation and the DSL sketch

The model has one organising idea: **every object belongs to exactly one organisation, and no relation may cross an organisation boundary except through a mandate**. Tenant isolation is therefore a property of the graph's shape, not a filter applied afterwards.

TYPE INVENTORY

Type	Key relations	Notes
organisation	principal, staff, accountant, field_agent, warden, support_admin	The tenancy boundary. Landlord account, management firm, society, hostel operator, vendor firm and the platform itself are all organisations.
mandate	grantor (org), grantee (org), scope, spend_ceiling, valid window	The only legal bridge between two organisations. Carries a validity condition, so an expired or revoked mandate stops granting without a job running.
property	organisation, owner, co_owner, managed_by (via mandate), assigned_agent	Permissions on a property cascade down to its units, beds, tickets and documents — deliberately, and only downward.
unit / room / bed	parent, occupant	Bed is a first-class object so a hostel resident's access is scoped to their bed, not the building.
occupancy	unit, tenant, co_tenant, guardian, payer	Lease, licence or commercial deed. Guardian gets payer and welfare relations without the occupant's full access.
invoice, payment, deposit, payout, statement	occupancy or property parent	Financial objects inherit visibility from their occupancy and property, never from organisation membership alone.
ticket, job	unit, reporter, assignee, vendor_org	A technician's access is a <i>conditional</i> relation valid only while the job is assigned and within its window.

listing, listing_media	property, public_viewer	The only types permitted a wildcard subject, and only when published.
enquiry, inspection	listing, prospect, assigned_agent	Gives a phone-verified prospect access to their own enquiry and nothing else.
document	parent object	Never independently permissioned; always inherits from what it evidences.
demo_session	demo organisation	Isolated exactly like a customer organisation; carries no relation to any real object and no administrative relation.

DSL SKETCH — THE SHAPE, NOT THE FINAL MODEL

```

type organisation
  relations
    define principal: [user]
    define staff: [user]
    define accountant: [user]
    define support_admin: [user with elevation_window]      # platform, time-bound, audited
    define can_administer: principal or support_admin

type mandate
  relations
    define grantor: [organisation]
    define grantee: [organisation]
    define active_manager: staff from grantee with mandate_valid  # condition: valid_from/valid_to

type property
  relations
    define org: [organisation]
    define owner: [user] or principal from org
    define mandate_ref: [mandate]
    define manager: active_manager from mandate_ref
    define field_agent: [user with assignment_valid]
    define can_view: owner or manager or field_agent or can_administer from org
    define can_manage_listing: owner or manager
    define can_approve_spend: owner or manager                # ceiling checked in the domain

type unit
  relations
    define parent: [property]
    define occupant: [user]
    define can_view: occupant or can_view from parent

type occupancy
  relations
    define unit_ref: [unit]
    define tenant: [user]
    define guardian: [user]
    define payer: tenant or guardian
    define can_view_financials: tenant or guardian or can_view from unit_ref

type job
  relations
    define ticket_ref: [ticket]
    define technician: [user with job_window]                # expires with the job
    define can_view: technician or can_view from ticket_ref

type listing
  relations
    define property_ref: [property]
    define published: [organisation]
    define public_view: [user:~] if published                 # the ONLY wildcard in the model
    define can_manage: can_manage_listing from property_ref

```

ISOLATION INVARIANTS — TESTABLE, AND TESTED

Invariant

Why it matters

Every object type resolves to exactly one organisation	An object with no organisation has no isolation boundary and cannot be reasoned about
No relation crosses organisations except through <code>mandate</code>	One bridge, one place to audit, one place to revoke
Permissions cascade downward only — property → unit → occupancy → financial object	Upward inheritance would let a tenant's access reach their neighbour's
Organisation membership alone grants nothing	A firm's accountant sees the properties they are assigned to, not everything the firm touches
<code>user:*</code> appears on published listings and their media, nowhere else	One wildcard, provable by inspecting the model rather than by testing every endpoint
Administrative relations are explicit, time-bound and audited	There is no implicit super-user, and elevation is visible
Time-bound relations expire by condition, not by cleanup job	A revoked mandate, an ended assignment or a finished job stops granting immediately
Every check is paired with database row-level security on the same request	Two independent mechanisms; a bug in either is not sufficient to leak data
Demo organisations hold no relation to any real object	The sandbox cannot become a path into customer data

HOW OPENFGA AND ROW-LEVEL SECURITY DIVIDE THE WORK

OpenFGA answers "may this subject take this action on this object". It is the decision point, and it understands relationships — ownership, mandates, assignments, occupancy, guardianship. **Row-level security answers "may this database session see rows belonging to this organisation".** It is the containment mechanism, and it understands nothing about intent. Both run on every request: the check decides, and the session binding — derived from the authorised organisation, never from a client-supplied value — contains. Neither is a substitute for the other, and the test suite proves each still denies with the other disabled.

10. Non-functional requirements

Area	Requirement
Money correctness	Currency stored as integer minor units with an explicit currency; floats prohibited in every layer including API payloads, exports and analytics. Balances always derived from immutable postings. A reconciliation job proves the ledger balances daily.
Reliability	The collection chain — invoice, reminder, collect, reconcile — is resumable at every step. Every multi-system operation is a durable workflow with compensations. Defined degraded mode: with the aggregator down, offline payments are still recorded and balances remain correct.
Idempotency	Every collection, payout, refund, dispatch and notification is idempotent. Every webhook is deduplicated on provider event id and is replay-safe. A retried operation can never double-charge, double-pay or double-notify.
Isolation	<code>tenant_id</code> on every table with database-enforced row-level security. Cross-organisation access always traverses an explicit, revocable grant. An automated isolation test suite runs on every pull request and blocks merge on failure.
Security	Fail-closed OpenFGA authorisation at the service boundary, independent of and in addition to database row-level security. Sensitive-action audit logging. Dependency, container and secret scanning in CI. Threat model maintained, with a security review gate for any story touching money, identity or documents.
Privacy	Aadhaar numbers never stored, logged or sent to analytics. PAN, bank and payout details encrypted or held in a secret manager. Telemetry excludes identifiers, document contents and full phone numbers. Retention matrix enforced automatically.
Performance	Budgets set against an entry-level Android on a 3G-equivalent connection: app cold start, payment screen interactive, ticket submit, and list scrolling. Reporting reads never touch the transactional primary.
Availability	Collection and payment paths carry the highest availability target. Rent due dates cluster at month start, so capacity is planned for a peak that is 8–10× the daily average, not for the average.
Recoverability	Stated RPO and RTO for the financial record, with a scheduled and evidenced restore drill. Point-in-time recovery must cover the window any financial dispute would need.
Observability	End-to-end tracing across the money path. Metrics on collection success, mandate health, reconciliation drift, dispatch time, SLA breach and notification delivery. Per-organisation support view. No PII in any span or log.
Auditability	Every state change attributable to an actor, with impersonation distinguishable from a user's own action. Financial history reconstructable as of any past date.
Accessibility	Stated WCAG level, keyboard and screen-reader support on web, adequate contrast and touch targets on mobile, and no critical action available only through colour or gesture.
Configurability	Statutory rates, deposit caps, fee models, SLA targets, liability matrices, reminder ladders and schedules of rates are configuration with an owner and a review date — never constants in code.
Scale assumptions	Design target: 500,000 units and beds, 1,000,000 monthly invoices, 250,000 monthly service jobs, with 60% of collections landing within a 72-hour window each month.

11. India regulatory summary

Each item below is a design constraint, not background reading. Every figure must be confirmed with counsel or a practising chartered accountant before the dependent feature ships — rates, slabs and thresholds change with every Budget and Council meeting, which is precisely why they live in versioned rule tables.

Area	Constraint on the product
UPI & autopay	UPI intent and collect for one-off payment; UPI Autopay mandates for recurring, subject to a per-mandate cap with a mandatory pre-debit notification; NACH for higher values. A missed pre-debit notification is a compliance failure, not a missed message.
Aggregator posture	RBI payment-aggregator directions bind how funds may be held and routed. dwellm8 holds no client funds; any vendor advance model is reviewed before build.
Agreements	The 11-month convention exists because 12 months or more attracts compulsory registration. Stamp duty is a state subject with different bases, several of which include a deposit component. E-stamping channels differ by state.
E-signature	Aadhaar-based eSign through a licensed provider produces a signed artefact and completion certificate. The Aadhaar number passes through the provider flow and is never retained by dwellm8. A non-Aadhaar path must exist, with its weaker evidentiary weight recorded.
Model Tenancy Act 2021	Adopted state by state. Where adopted it caps residential deposits, requires written agreements filed with a Rent Authority, and defines notice and eviction process. Modelled as a per-state flag with the cap enforced at agreement time.
Tenant verification	Police tenant verification is an owner obligation in several states, with the channel differing by state. Tracked as a compliance task with evidence; the platform states plainly whose duty it is.
GST	Residential rent to an unregistered individual is exempt; renting a residential dwelling to a registered person attracts reverse charge in defined cases; commercial rent is taxable; management fees and platform subscription are taxable services; RWA maintenance is exempt up to a per-member monthly threshold evaluated <i>per member</i> , not per unit. Hostel and PG supply raises composite-supply questions that must be settled with advice.
TDS on rent	§194-I for business deductors, §194-IB for individuals above the monthly threshold with Form 26QC and 16C, §195 for any tenant of a non-resident landlord with no small-value exemption. The path follows from deductor class and landlord residency, both captured at tenancy creation.
TDS on services	§194-C applies to vendor payments above threshold; vendor GST registration status determines invoice treatment. Both are handled in the payout engine, not left to the manager.
DPDP Act 2023	Purpose-limited collection with notice in English and at minimum Hindi, consent as a recorded artefact, implemented rights endpoints, a retention matrix that reconciles erasure with statutory retention, and a breach notification runbook.
RERA	Binds listings, agent conduct and advertising claims. Where dwellm8 or its agents originate leases or sales, agent registration and claim substantiation apply.
Society bye-laws & Co-op Acts	Govern maintenance dues, interest on arrears, non-occupancy charges, NOC for sale and transfer. Per-society configuration, never platform constants.
Sale & registration	Stamp duty and registration fee by state, TDS under §194-IA on consideration above the threshold, and higher rates for non-resident sellers. Society NOC and dues clearance gate a resale.

12. Phased delivery plan

The order below is chosen so that each phase is independently sellable, and so that the pieces that cannot be retrofitted — the ledger, the unit model with beds, multi-tenancy and the cost-sharing engine — are built before anything depends on them.

Phase	Window	Ships	Sellable to	Proves
P0	W0–W6	Foundations — tenancy model, ledger design, unit/room/bed model, auth, cost-sharing rule design, platform-fee and bearer model, Admin console skeleton, CI/CD, threat model, privacy posture	Nobody (internal)	That the irreversible decisions are made deliberately
P1	M1–M3	Residential core — properties, tenancies, invoicing, UPI collection, offline payments, receipts, reminder ladder, ledger, reconciliation, basic tickets, owner dashboard	Individual landlords	That collection actually works and reconciles
P2	M3–M6	Tenant app, agreements with stamping and eSign, KYC, deposits and move-out settlement, autopay mandates, maintenance with liability matrix	Landlords + their tenants	That the tenant becomes a user, not just a payer
P3	M6–M9	Manager platform — portfolios, staff, mandates and spend authority, owner payouts, statements, fees, GST and TDS, manager console (web)	Property management firms	That the middle layer is the scale channel
P4	M9–M12	Service marketplace — vendor onboarding and verification, dispatch, technician app, quotes and approvals, job costing, vendor payouts, ratings	All three, plus vendors	That maintenance becomes revenue, not cost
P5	M12–M15	Hostel vertical — bed inventory and allocation, guardian payer and visibility, mess and leave, attendance and in–out, warden console, hostel analytics	Hostel and PG operators	That one core carries a second vertical cheaply
P6	M15–M18	Commercial vertical — area-based rent, CAM with true-up, metered utilities, escalation ladders, registration path, fit-out and compliance tracking	Commercial landlords and managers	That high-value tenancies are billable correctly
P7	M18–M24	Transact — rental listings and lead pipeline, site visits, lease origination; then property sale, due-diligence pack, registration and brokerage	Owners, buyers, agents	That operating history converts into transaction value
P8	M24+	Scale — premium AI tier (₹499/month) , whitelabel, enterprise SSO, analytics and benchmarks, AMC packages, partner ecosystem	Enterprise	That the platform becomes infrastructure

WHY THE MANAGER COMES BEFORE THE MARKETPLACE

The service marketplace is the highest-margin part of this product, and it is tempting to build it early. It should not be. A marketplace with no job volume cannot attract good vendors, and good vendors are the only thing that makes it work. The property manager brings hundreds of units — and therefore hundreds of monthly tickets — in a single onboarding. Building the manager platform first (P3) creates the demand density that makes P4 viable on day one. Reversing the two produces an empty marketplace and a churned vendor base that is very hard to win back.

13. Assumptions and open decisions

11.1 Assumptions this document makes

1. Launch is India-only, in a small number of states chosen by e-stamping and verification coverage rather than by market size.
2. One identity provider serves everyone; there is no separate staff identity stack, and platform administrators hold no implicit privilege.
3. **There are no subscriptions.** The entire product is free and monetised by a flat 2.99% fee on payments transacted through the app; the premium AI tier at ₹499 per organisation per month is the only paid plan and arrives later.
4. Aggregator cost is netted from the 2.99%, so margin varies by payment instrument and must be monitored per instrument from launch.
5. AI is never load-bearing: a non-paying organisation can operate a property end to end with no blocked action.
6. dwellm8 never holds client funds. Every design here assumes settlement through a regulated aggregator.
7. Residential is the beachhead. Hostel and commercial reuse the same core and are sequenced after the manager platform exists.
8. Sales is an extension of an operating relationship, not a standalone portal business. dwellm8 will not win a listings-portal fight and does not attempt one.
9. Technicians are partner-employed. The platform must not drift into a de facto employment relationship.
0. WhatsApp remains an available and economically viable primary channel; SMS fallback is proven before launch.

11.2 Decisions required before build

Decision	Why it blocks	Needed by
Launch states	Determines e-stamping channel, stamp duty rules, police verification flow, Model Tenancy Act applicability	P0
Mobile framework — decided	React Native on Expo with expo-router and NativeWind, matching mark8ly and HomeChef (§9.3)	Closed
Auth provider — decided	Google Identity Platform for every user including admin; OpenFGA plus RBAC for authorisation (§9.4)	Closed
OpenFGA authorisation model	The type and relation definitions, and the check-latency and caching strategy on the critical path	P0
Payment aggregator and payout rails	Mandate caps, payout timing, settlement format and the vendor-payment model all follow from it	P0
Whether the manager may collect into their own account	Changes the entire money-flow, custody and reconciliation design	P0
Vendor payment model	Platform-advanced-then-recovered versus direct-to-vendor has regulatory and working-capital consequences	P3
Hostel GST treatment	Composite supply of stay plus food materially changes billing; requires professional advice	P5
Who bears the 2.99% fee — decided	The property manager, generalised as "the party earning the operating margin"; owner where self-managed, operator for hostels, society for dues, vendor for jobs (§5.5)	Closed

Fee base, cap and tiering	Whether the fee stays on gross rent at a flat rate, or moves to the manager's fee, a per-transaction cap or a volume tier — all rule-table values, but the market answer is unknown	P3
Fee treatment on cards and NACH	Aggregator cost on some instruments approaches or exceeds 2.99%; capping, surcharging or restricting an instrument must be decided	P1
GST treatment of the platform fee	The fee is a taxable service; whether it is charged on top of or within 2.99% affects both pricing and every invoice	P0
AI tier scope and cost ceiling	Which capabilities sit behind ₹499 and what per-organisation inference cost is sustainable at that price	P8
Marketplace fee application	Whether the 2.99% on a job is taken from the vendor, added for the customer, or absorbed in the job price	P4
Brokerage and RERA posture	Whether dwellm8 acts as an agent or only as a facilitator determines registration obligations	P7
Insurance and liability	Who is liable when a dispatched technician causes damage in a customer's home	P4

11.3 Principal risks

Risk	Mitigation the backlog must carry
The 2.99% fee drives users back to cash	Bearer setting configurable and disclosed; value must be visibly worth the fee (documents, maintenance, ledger, compliance); offline recording stays free so the ledger stays true even when the rail is bypassed
Payment mix erodes margin below the fee	Instrument-level margin monitoring from launch; UPI-first design; explicit decision on cards and NACH
Free product with no subscription floor cannot fund support	Support cost per organisation tracked against fee revenue; self-service and the Admin console reduce human touch
Owners churn after one collection failure	Reconciliation, alerting and a degraded mode that still shows correct balances
Marketplace launches without vendor density	Sequence the manager platform first so job volume exists on day one; launch category-by-category and city-by-city
A dispatched technician causes damage or harm	Verification of every technician, defined liability and insurance posture, OTP-gated entry and completion, evidence capture
Deposit and cost-split disputes become support load	Rules decided in advance, itemised evidence, tenant acknowledgement steps, and a defined arbitration route
Compliance drift across states	State-scoped rule tables with an owner and a review date; never a hardcoded constant
Hostel operators refuse to migrate from whiteboards	Import tooling, a parallel-run mode, and a bed-allocation flow faster than the whiteboard it replaces
KYC data becomes a liability	Never store Aadhaar; result-only retention; encryption, access audit and a retention matrix
The product becomes three products	One core enforced architecturally — verticals add modules and billing bases, never a parallel data model

14. Appendix — competitive parity check

dwelIm8 was compared against PropertyMe, Australia's largest property-management platform and the most mature example of the manager-in-the-middle model, in July 2026. The purpose was to find capability gaps, not to copy a pricing or regulatory model that does not apply in India.

14.1 What their model tells us

Dimension	PropertyMe (Australia)	dwelIm8 (India)
Charging model	Per-property subscription to the agency, roughly A\$1.50–2.15 per property per month by plan, payments an add-on at about A\$1.10 per property. No percentage of rent.	No subscription. 2.99% platform fee deducted once, by the manager, at payout.
Effective share of rent	On the order of 0.1%	2.99% — about 30–37% of a manager's own 8–10% fee
Why the difference is defensible	Australian management is performed by licensed agencies operating statutory trust accounts; the software is a regulated ledger sold as infrastructure, and the percentage economics sit with the agency. India has no equivalent regime, managers are far smaller, subscription willingness-to-pay is low, and dwelIm8 bundles collections, maintenance, a vendor marketplace, documents and compliance rather than selling a ledger. A transaction model can be right here where a subscription is not — provided rate, base and cap remain rule-table values.	
Tenant-side charging	None — tenants pay by BPAY, direct debit or EFT at no cost. Card surcharging is being banned outright from 1 October 2026 following the RBA's retail payments review.	None. The tenant pays exactly the rent; instrument costs are passed through separately and UPI carries none.
Where the fee appears	Agency's technology and admin cost, passed to the owner as a statement line	Platform fee, itemised on the owner statement beside the management fee

14.2 Capability gaps found, and now specified

Twenty-five stories were added as a result of this comparison. Where a capability was Australia-specific — statutory trust accounting, sales trust, BPAY batching — it was translated into its Indian equivalent rather than copied.

Capability they have	Our status before	Now specified as
Automatic bank feeds and statement import	Missing	BIL-18, BIL-19 — daily feeds plus import for banks without one. The most important gap: most Indian rent moves bank-to-bank, and we only reconciled the gateway.
Routine periodic inspections with photos, notes and a signed report	Missing — we had move-in, move-out and prospect viewings only	MNT-14, MNT-17
Inspection route planning from geodata	Missing	MNT-15
Tenant-completed self inspection	Missing	MNT-16

Compliance register — smoke alarms, pool safety, insurance renewals	Partial — expiry tracking only	MNT-18, translated to fire safety, lift, electrical, water tank, insurance and society NOCs
Supplier bill capture, scanning and allocation	Missing	VND-17
Recurring supplier bills and outgoings	Missing	VND-18 — society dues, AMC, insurance, property tax
Multi-step bill approval	Partial	VND-19
Disbursements triggered by date or balance	Partial — fixed schedule only	FIN-18
Bulk disbursement and bank batch files	Missing	FIN-19
Period close and audit-ready pack	Missing	FIN-20
Automated fee types — letting, admin, renewal	Partial — management fee only	BIL-21
Rent credits and concessions	Missing	BIL-20
Key and access-card management	Missing	INV-14
Message template library with merge fields	Partial — fixed ladder only	NTF-10
Inbound replies into a unified inbox	Missing	NTF-11
Activity feed on every record	Missing — we had an audit trail, not a readable history	NTF-12
Internal mentions and hand-off	Missing	NTF-13
Portal syndication	Explicitly excluded	SAL-33 — reinstated at P3 for the Indian portals
Portfolio KPIs — gained and lost, response times, overdue	Partial	RPT-11
Property view versus client view	Missing	RPT-12
Workflow automation and a no-code builder	Missing	Module M20 — AUT-01 to AUT-05
In-context compliance guidance (AI)	Partial	AIX-16

14.3 Where dwellm8 goes further

- **Three verticals on one core** — hostel and PG with bed-level inventory and a guardian payer, and commercial with CAM and metered utilities, alongside residential. PropertyMe is residential and commercial only, with no bed-level model.
- **An embedded service marketplace** — verified vendors, OTP-gated jobs, quotes, warranty and settlement — rather than integrating with a third-party job service.

- **Cost-sharing as a rule engine** — who pays is decided from the agreement before dispatch, rather than being a note on a work order.
- **Public discovery and inspection booking** in the same product that operates the tenancy, so a listing converts into a tenancy with no re-keying.
- **India's compliance mechanics as product** — e-stamping, Aadhaar e-sign, police verification, GST and TDS as workflow rather than advice.
- **A no-sign-up sandbox** for evaluation and for staff training on releases.

DOCUMENT CONTROL

Version 2.0 sets the fee at 2.99%, fixes the backend as one modular monolith API serving every client, adds 25 capabilities found by benchmarking against PropertyMe, and adds module M20 workflow automation. Version 1.9 charged the platform fee once, deducted by the manager at payout, with instrument costs passed through separately. Version 1.7 settled the fee bearer, specifies the authorisation model and makes dwellm8 Ops a full web console as well as an app. Version 1.6 added module M19 — a no-sign-up sandbox anyone can use before creating an account, and a training and release-preview sandbox for existing customers and their staff. Version 1.5 defined the five access tiers, the no-anonymous rule for administrative surfaces, and the demonstration-data workspace. Version 1.4 settled identity and authorisation — Google Identity Platform for every user, OpenFGA plus RBAC for permissions. Version 1.3 added public discovery, listings and inspection scheduling, and settles the technology stack. Version 1.2 made the Admin surface web and mobile together. Version 1.1 replaced the subscription model with a flat 2.99% transaction fee and adds the premium AI tier, the Admin web console, the end-to-end workflows and the integration framework. This document defines **scope and intent at the epic level**. It does not replace the engineering backlog: every story here decomposes into engineering stories on the delivery board using the standard template, each with testable acceptance criteria, a mandatory failure scenario and the platform guardrails. Where this document and the repository's existing product brief differ on vertical scope, **this document supersedes it** and the brief will be updated to match.